

Student Portal & Hostel Services System

PROJECT 1 - Complete Service Documentation

40	6	7	5
Services	Modules	Teams	Testers

Document Overview
Detailed specifications for all 40 services across Authentication, Course Management, Academic Records, Assignment Portal, Hostel Management and Complaint System. The document also includes the Testing Team and provides a consistent description, service metadata, and UML diagrams for every service.
Team Leaders: Kairav Kumar, Ayush Wani, Durgesh Shelke, Koppiseti Charvi Sri
CS 345 / CS 346 - Prof. Pradeep K. Das
Software Engineering Project - Student-Side System

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Testing Team

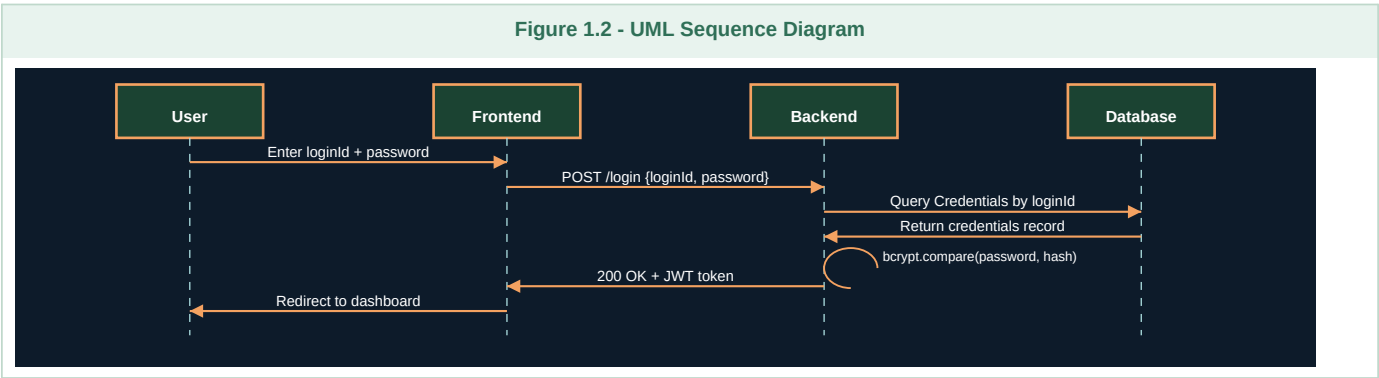
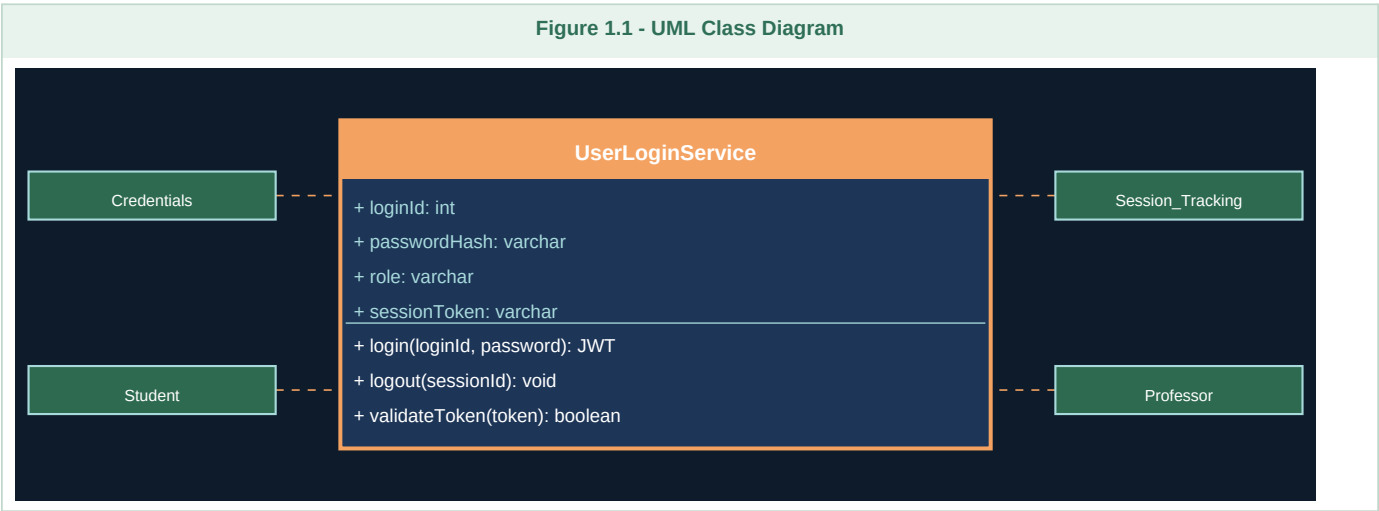
- Service 36: IntegrationTestService - chokkaku venkata yaswanth
- Service 37: UnitTestService - praveen kumar kanchari
- Service 38: PerformanceTestService - Sachin Kumar Tanti
- Service 39: SecurityTestService - Shravya Lanka
- Service 40: RegressionTestService - Shrey Chandra

Module 1: Authentication

Members: Deepit Gupta, Lakshay Bansal, Raghav Goyal, Vaibhav Singh, YEREDLA KOUSHIK REDDY

Service 1: UserLoginService [Deepit Gupta]

Description	
Authenticates users (Student/Professor/Admin) by verifying institutional credentials against the secure database and issuing a JWT token upon success.	
Attributes	Methods
loginId: int passwordHash: varchar role: varchar sessionToken: varchar	login(loginId, password): JWT logout(sessionId): void validateToken(token): boolean
Related Classes	
Credentials Session_Tracking Student Professor	

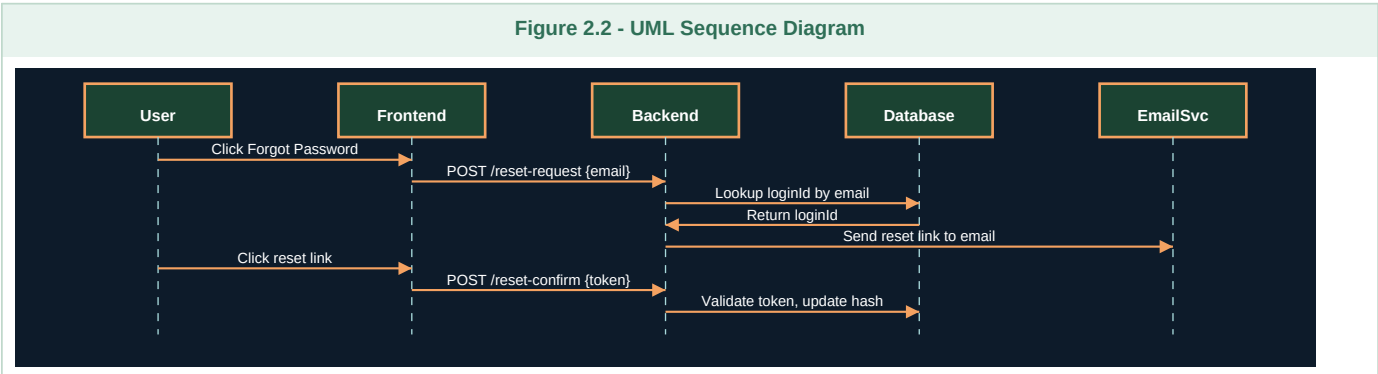
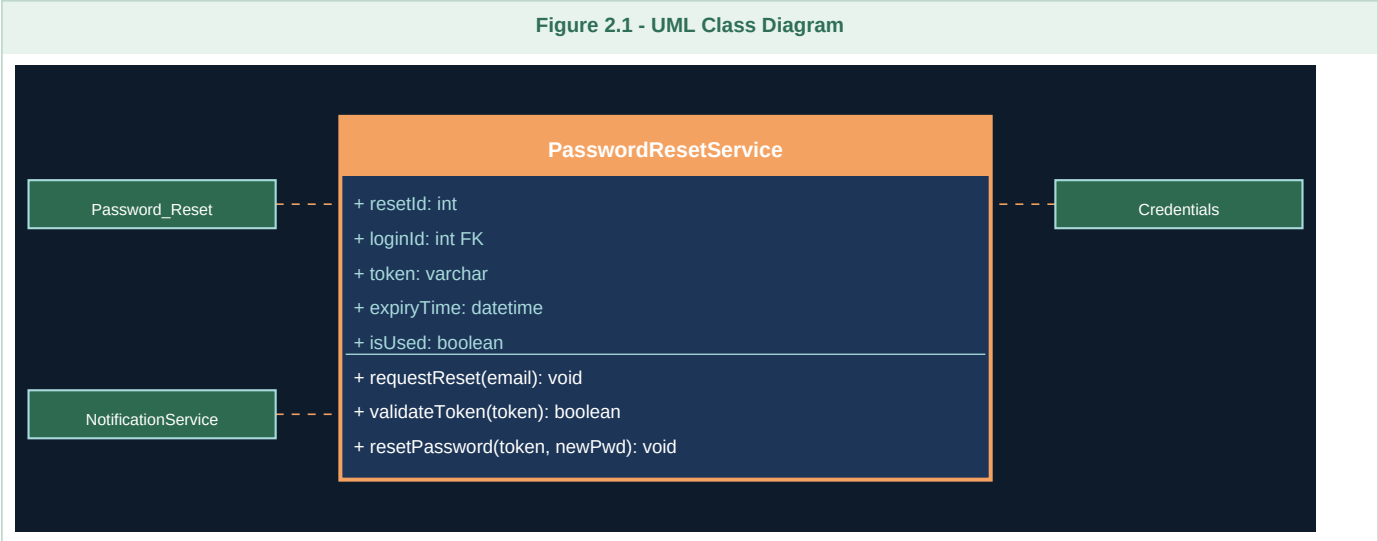


Service 2: PasswordResetService [Lakshay Bansal]

Description
Handles forgot-password flow: generates a time-sensitive single-use reset token, dispatches it to the registered institutional email, and validates it upon use.

Attributes	Methods
resetId: int loginId: int FK token: varchar expiryTime: datetime isUsed: boolean	requestReset(email): void validateToken(token): boolean resetPassword(token, newPwd): void

Related Classes
Password_Reset Credentials NotificationService



Service 3: SessionManagementService [Raghav Goyal]

Description
Tracks active user sessions across devices, enforces a 30-minute inactivity timeout, and provides session revocation for logout or suspicious activity.

Attributes	Methods
sessionId: int PK loginId: int FK token: varchar deviceInfo: varchar ipAddress: varchar expiresAt: datetime	createSession(loginId, device, ip): Session revokeSession(sessionId): void checkInactivity(sessionId): boolean listActiveSessions(loginId): Session[]

Related Classes
Session_Tracking Credentials

Figure 3.1 - UML Class Diagram

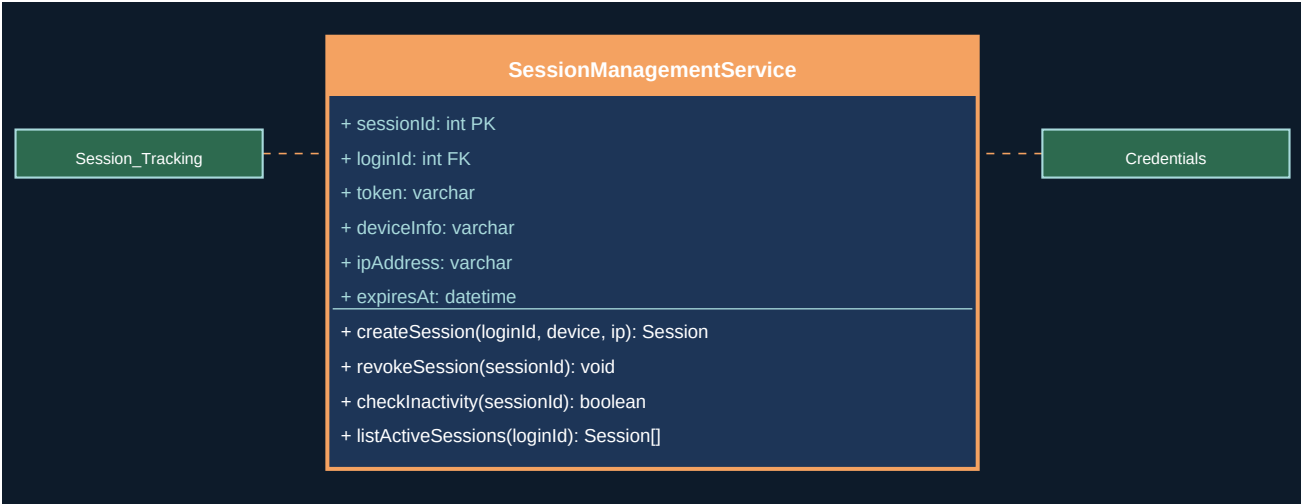
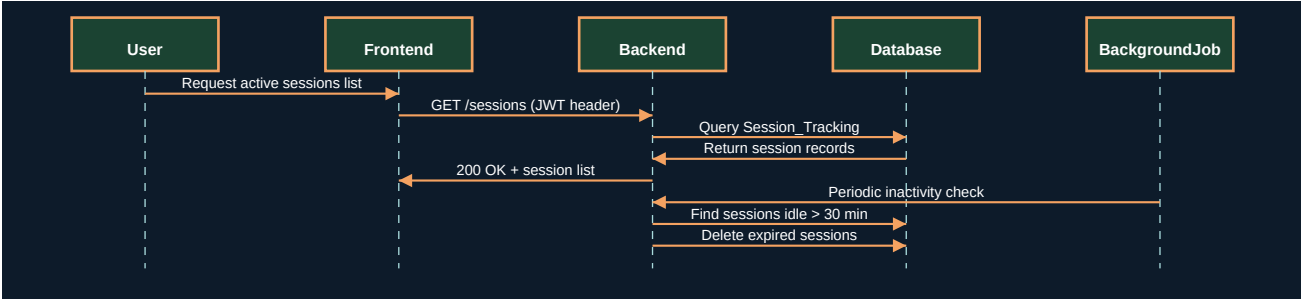


Figure 3.2 - UML Sequence Diagram



Service 4: AccountLockoutService [Vaibhav Singh]

Description

Monitors failed login attempts per account and automatically locks the account for 15 minutes after 5 consecutive failures, preventing brute-force attacks.

Attributes	Methods
loginId: int failCount: int lockedUntil: datetime lastAttempt: datetime	recordFailedAttempt(loginId): void isLocked(loginId): boolean unlockAccount(loginId): void getFailCount(loginId): int

Related Classes

Credentials | AuditLog | Admin_Staff

Figure 4.1 - UML Class Diagram

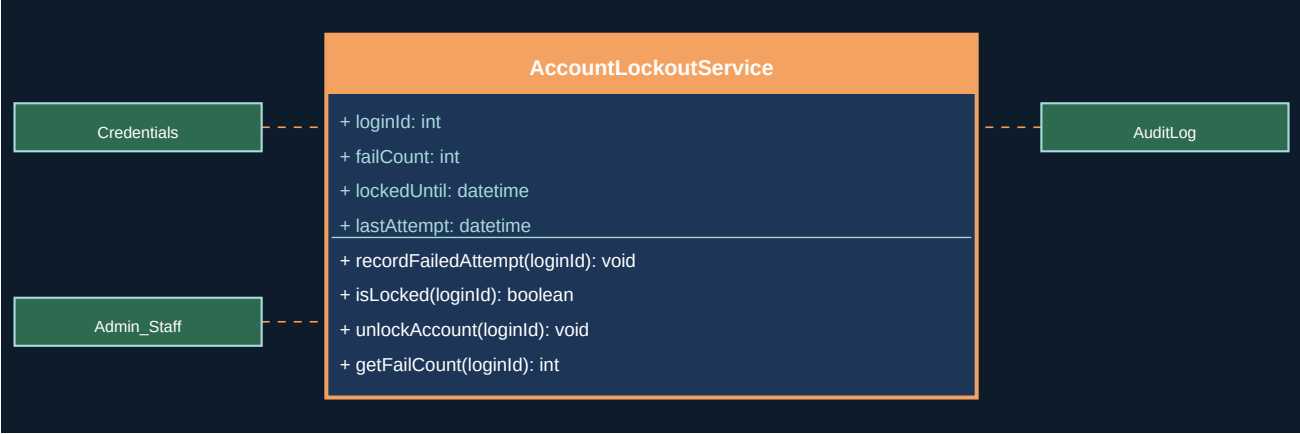
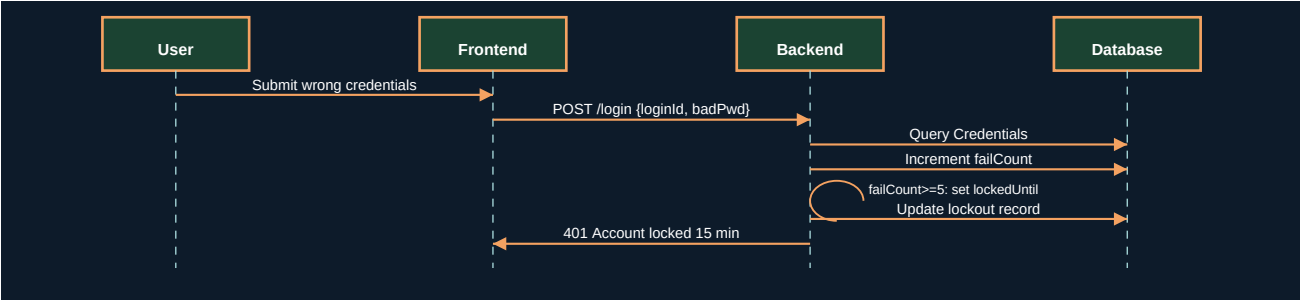


Figure 4.2 - UML Sequence Diagram



Service 5: RBACService [YEREDLA Koushik Reddy]

Description

Implements Role-Based Access Control: dynamically grants or restricts access to portal modules based on the authenticated user's role embedded in the JWT.

Attributes	Methods
<code>loginId: int</code> <code>role: varchar</code> <code>permissions: string[]</code> <code>resource: varchar</code>	<code>assignRole(loginId, role): void</code> <code>checkPermission(token, resource): boolean</code> <code>getRoles(loginId): string[]</code> <code>revokeRole(loginId, role): void</code>

Related Classes

Credentials | Student | Professor | Admin_Staff

Figure 5.1 - UML Class Diagram

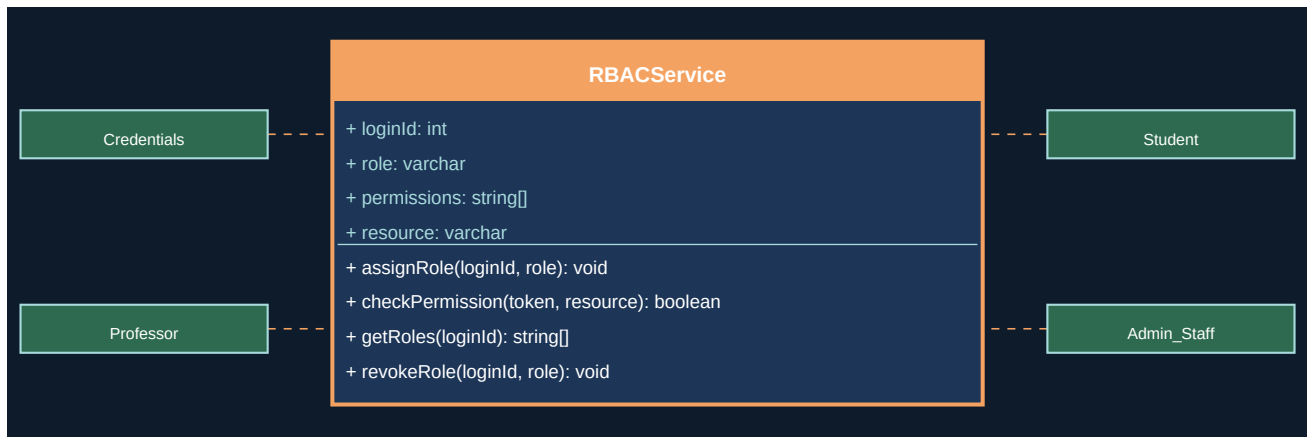
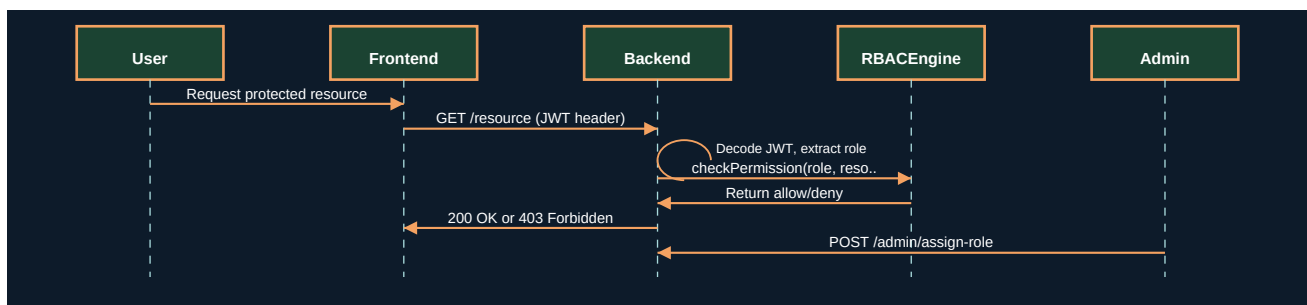


Figure 5.2 - UML Sequence Diagram

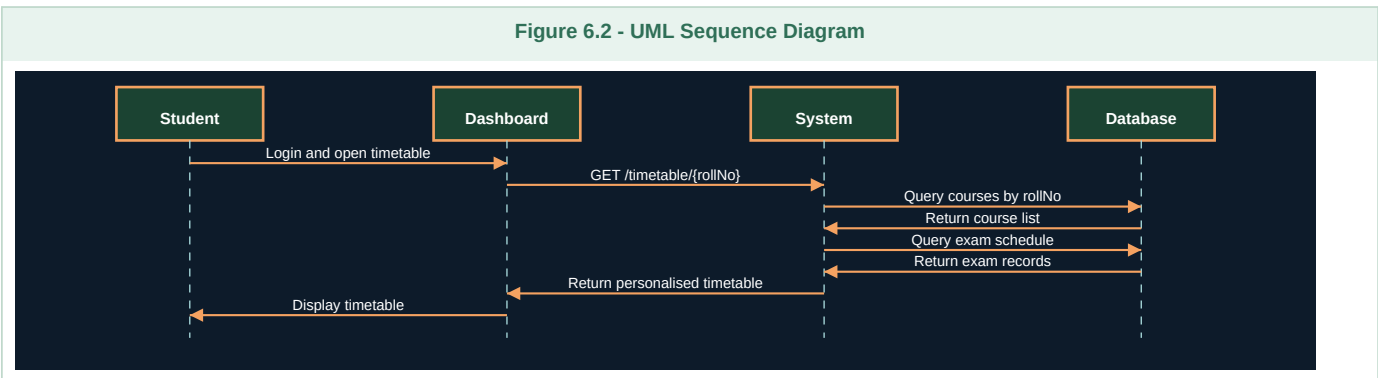
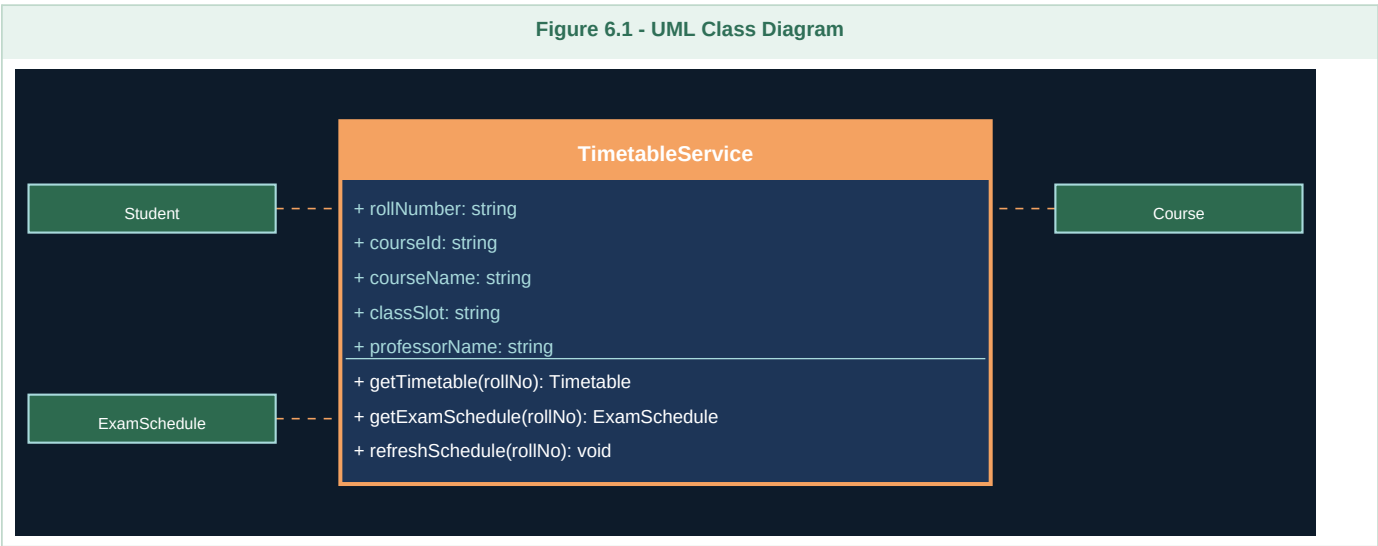


Module 2: Course Management

Members: Abhi Tanwar, Aeyaz Adil, Atharv Agarwal, Garima Kalra, Maddala Bogadi Sushanth

Service 6: TimetableService [Abhi Tanwar]

Description	
Dynamically generates and displays a personalised timetable for each student based on their active Roll Number, reflecting real-time admin changes to course assignments.	
Attributes	Methods
rollNumber: string courseId: string courseName: string classSlot: string professorName: string	getTimetable(rollNo): Timetable getExamSchedule(rollNo): ExamSchedule refreshSchedule(rollNo): void
Related Classes	
Student Course ExamSchedule	

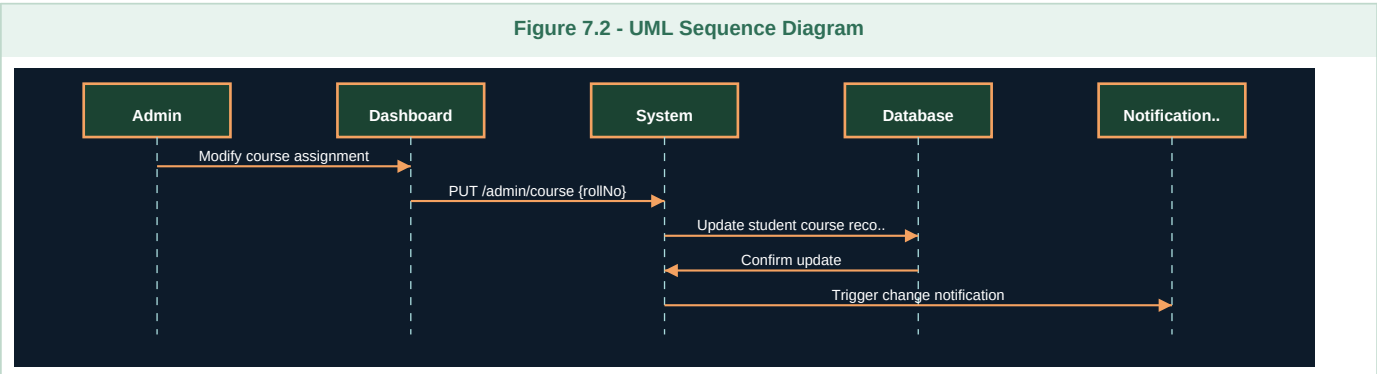
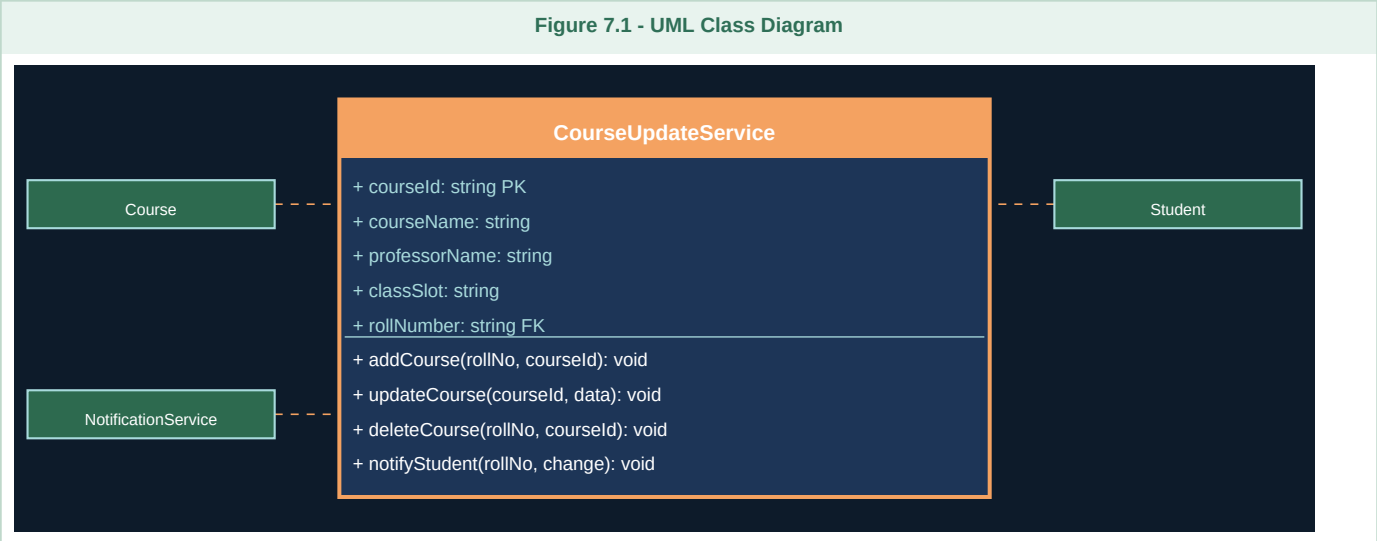


Service 7: CourseUpdateService [Aeyaz Adil]

Description
Allows admins to add, modify, or delete course assignments for students; propagates changes in real-time to the student's personal dashboard via notifications.

Attributes	Methods
courseId: string PK courseName: string professorName: string classSlot: string rollNumber: string FK	addCourse(rollNo, courseId): void updateCourse(courseId, data): void deleteCourse(rollNo, courseId): void notifyStudent(rollNo, change): void

Related Classes
Course Student NotificationService



Service 8: DataChangeRequestService [Atharv Agarwal]

Description
Manages student-initiated field-level data change requests (e.g., address, contact). Stores old/new values with proof documents, routes to admin for approval, and notifies the student.

Attributes	Methods
requestId: int PK rollNumber: string FK fieldName: string oldValue: text newValue: text status: string	submitRequest(rollNo, field, newValue, proof): RequestId reviewRequest(requestId, decision): void getRequestStatus(requestId): Status

Related Classes
DataChangeLog Student Admin Vault

Figure 8.1 - UML Class Diagram

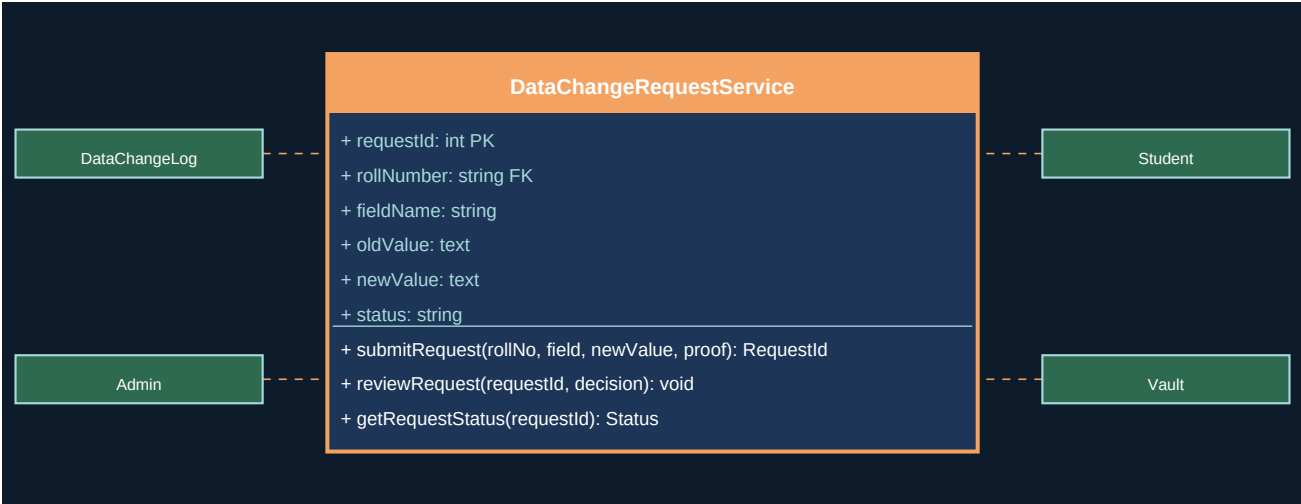
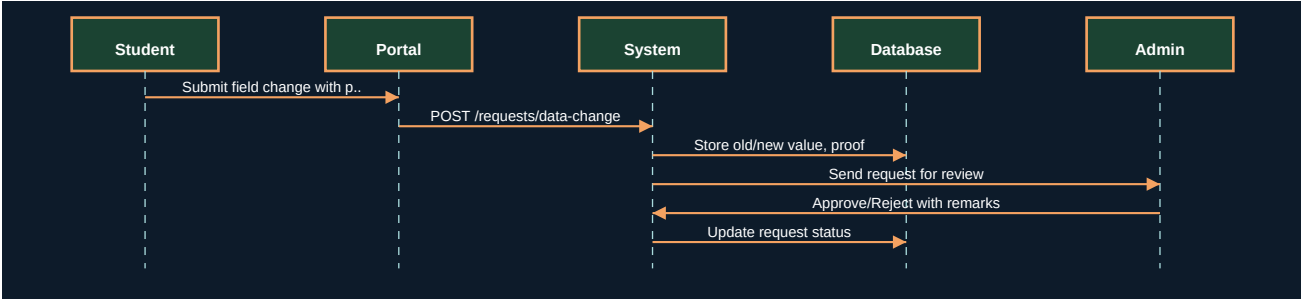


Figure 8.2 - UML Sequence Diagram



Service 9: CertificateRequestService [Garima Kalra]

Description

Handles Bonafide Certificate and NOC (Passport/Internship) requests with full workflow tracking. Admin applies digital signature on approval; student downloads the certificate.

Attributes	Methods
requestId: int PK rollNumber: string FK certificateType: string status: string documentUrl: string	requestCertificate(rollNo, type): RequestId approveCertificate(requestId, adminId): void downloadCertificate(requestId): File trackStatus(requestId): Status

Related Classes

CertificateRequest | Vault | Admin | Student

Figure 9.1 - UML Class Diagram

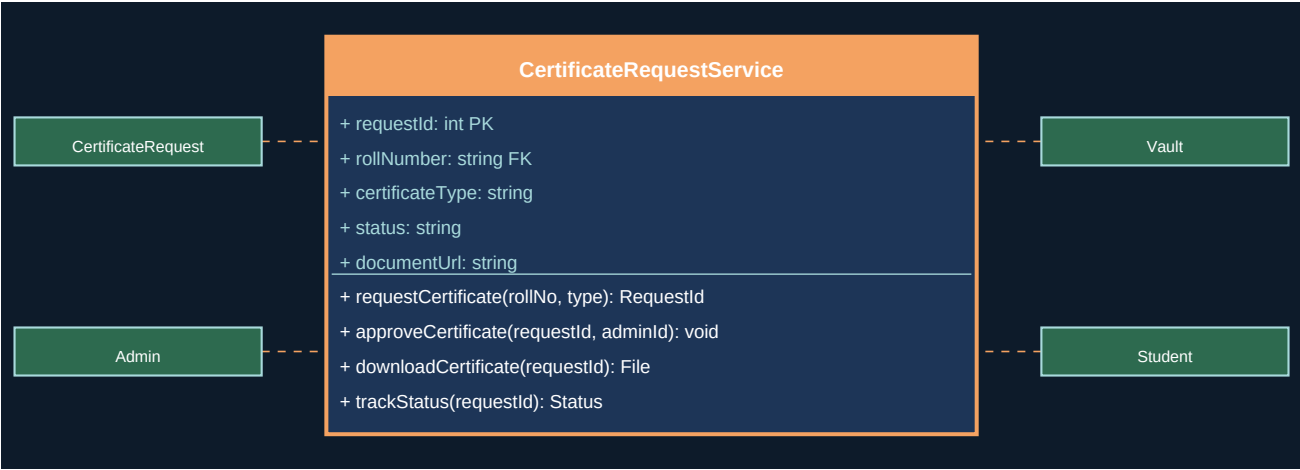
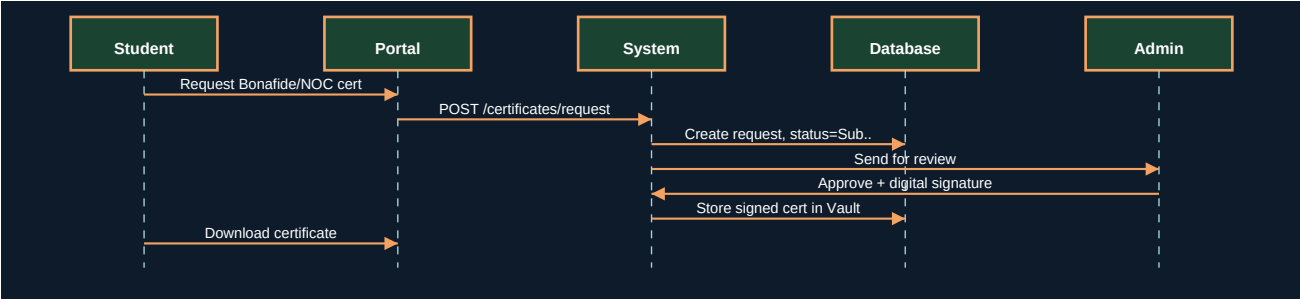


Figure 9.2 - UML Sequence Diagram



Service 10: IDCardService [Maddala Bogadi Sushanth]

Description
Provides ID card renewal and replacement workflows. Renewal requires FIR proof; replacement requires old and new details. Integrates with the fee gateway to confirm payment before dispatch.

Attributes	Methods
requestId: int PK rollNumber: string FK requestType: string transactionId: string dispatchStatus: string	requestRenewal(rollNo, firProof): RequestId requestReplacement(rollNo, details): RequestId verifyPayment(transactionId): boolean trackDispatch(requestId): Status

Related Classes
IDCardRequest FinanceSection Admin PaymentGateway

Figure 10.1 - UML Class Diagram

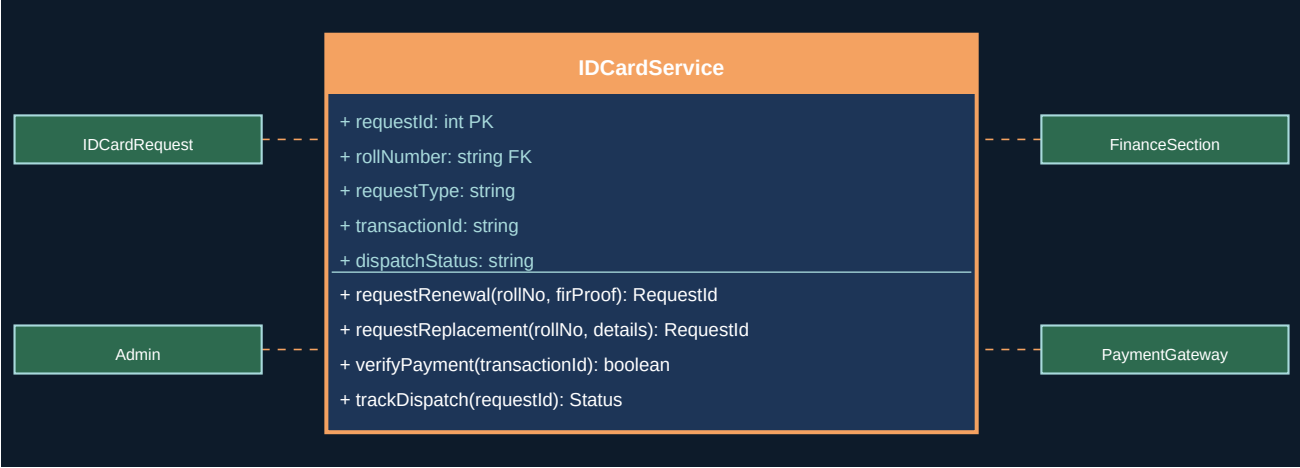
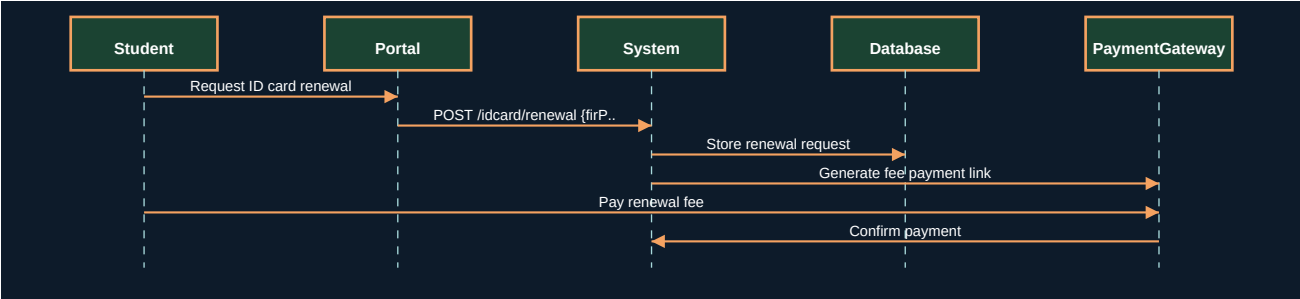


Figure 10.2 - UML Sequence Diagram



Service 11: FeePaymentService [Module 2 - Payment]

Description

Provides an integrated payment gateway for ID card fees and event registrations. Validates transaction IDs with the Finance Section before marking payments as confirmed.

Attributes

paymentId: string PK
userId: string FK
amount: decimal
transactionId: string
status: string
paidAt: datetime

Methods

initiatePayment(userId, amount, purpose): PaymentLink
confirmPayment(transactionId): boolean
validateWithFinance(transactionId): boolean
getPaymentHistory(userId): Payment[]

Related Classes

Payment | FinanceSection | IDCardRequest | EventRegistration

Figure 11.1 - UML Class Diagram

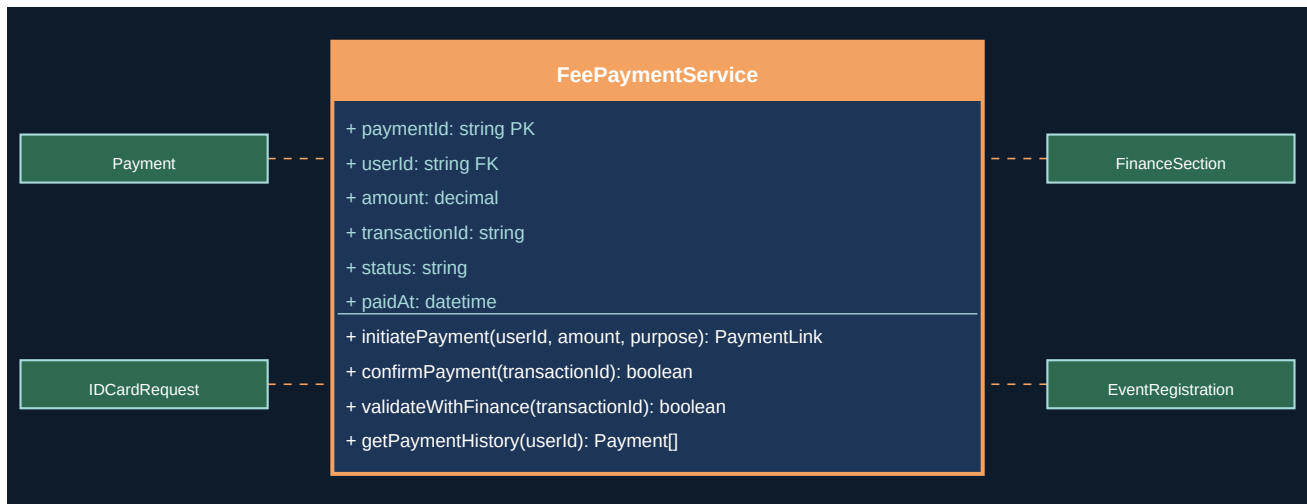
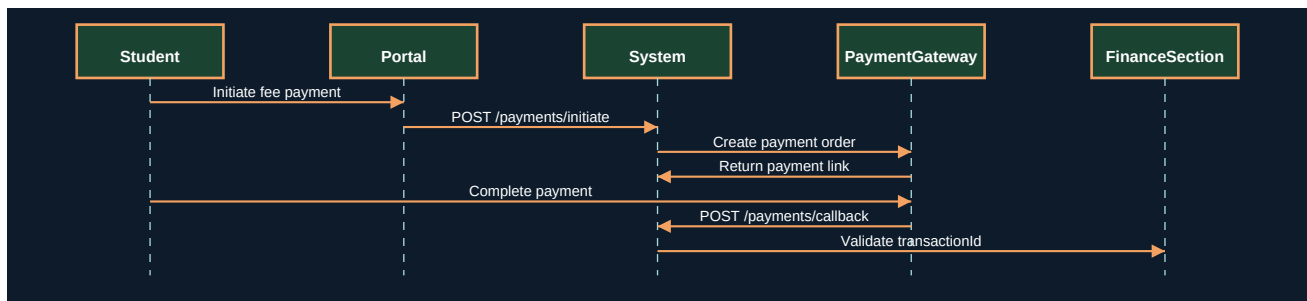


Figure 11.2 - UML Sequence Diagram

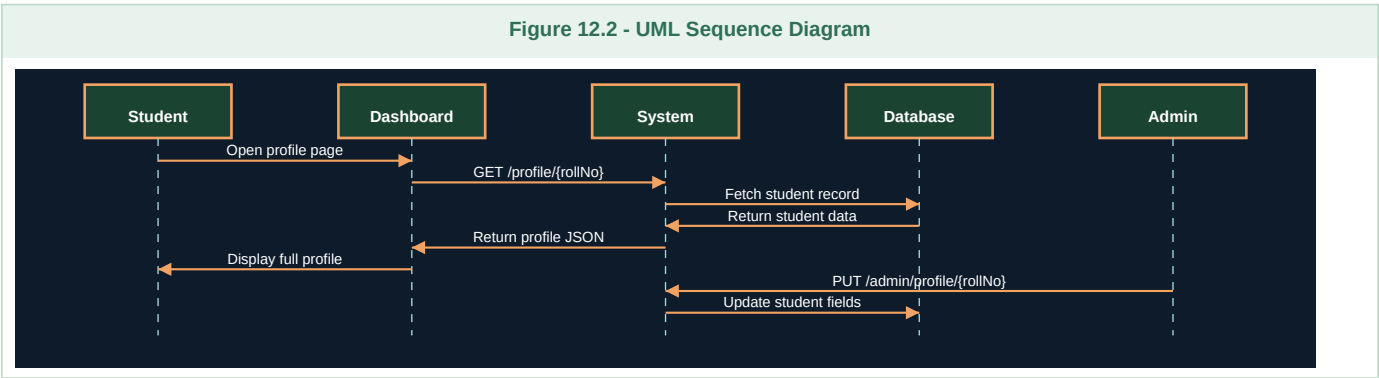
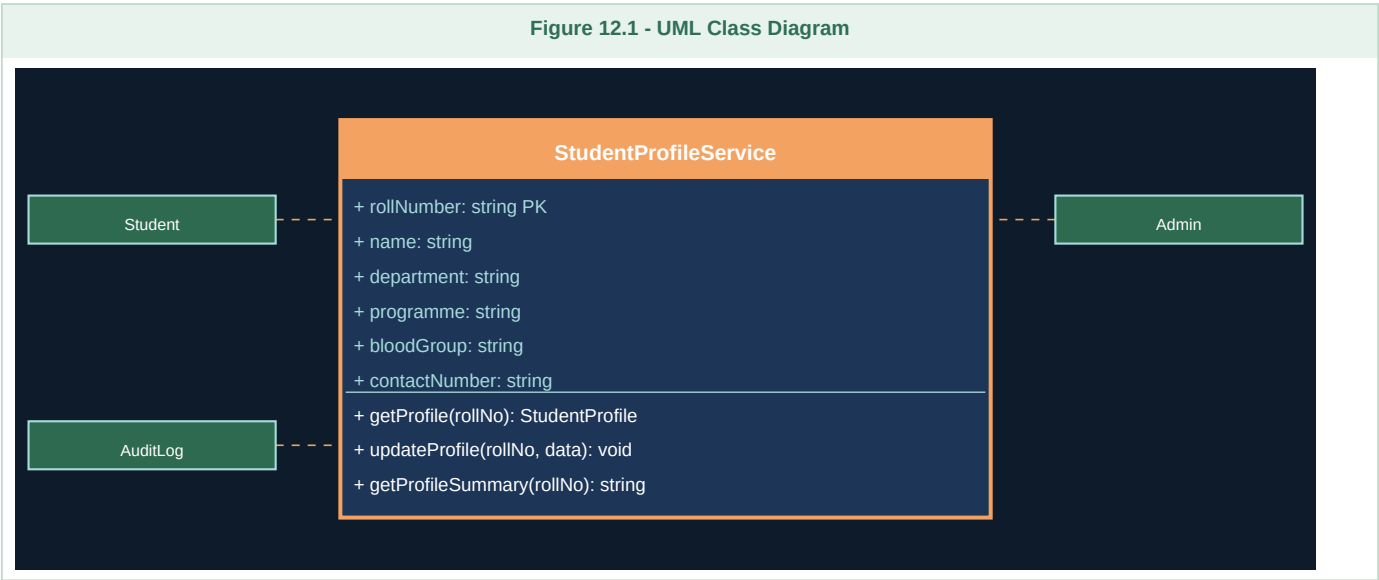


Module 3: Academic Records

Members: Aditya Anand, Divyansh Malviya, Harsh, Nakshatra Kumar, Tarandeep Kaur

Service 12: StudentProfileService [Aditya Anand]

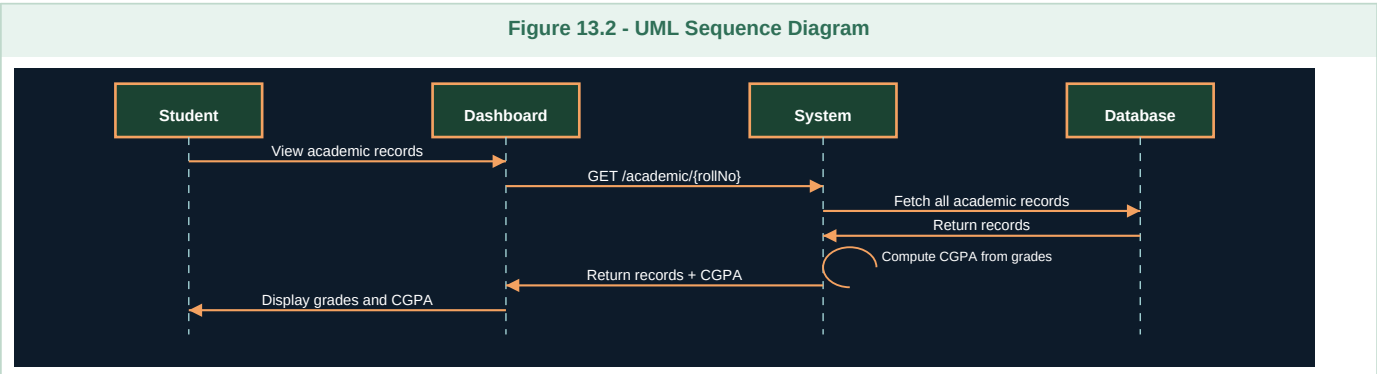
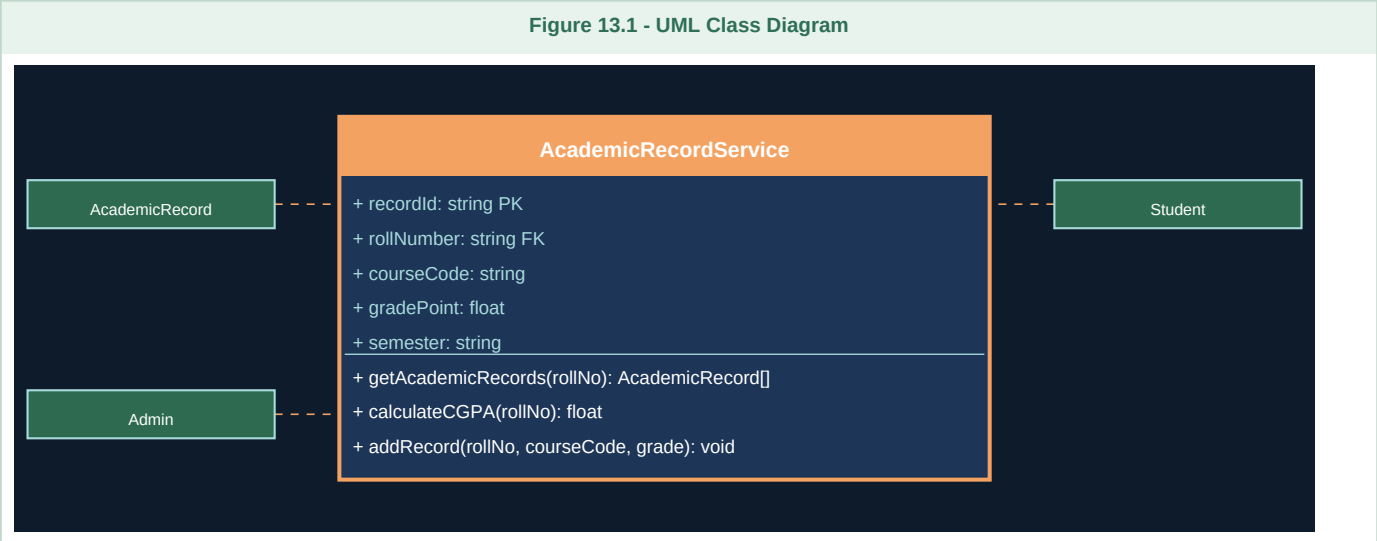
Description	
Manages the complete student profile including personal details, department, programme, and emergency contacts. Exposes a read-only view to students and an update interface to admins.	
Attributes	Methods
rollNumber: string PK name: string department: string programme: string bloodGroup: string contactNumber: string	getProfile(rollNo): StudentProfile updateProfile(rollNo, data): void getProfileSummary(rollNo): string
Related Classes	
Student Admin AuditLog	



Service 13: AcademicRecordService [Divyansh Malviya]

Description
Provides students access to their academic records including course codes, grades, and CGPA computation. Records are managed by admin and visible in read-only mode to students.

Attributes	Methods
recordId: string PK rollNumber: string FK courseCode: string gradePoint: float semester: string	getAcademicRecords(rollNo): AcademicRecord[] calculateCGPA(rollNo): float addRecord(rollNo, courseCode, grade): void
Related Classes	
AcademicRecord Student Admin	



Service 14: ExamScheduleService [Harsh]

Description	
Manages and displays personalised exam schedules per student, including exam type, date, room allocation, and course. Syncs with admin updates in real-time.	
Attributes	Methods
examId: string PK courseId: string FK examType: string examDay: date examRoomNumber: string	getExamSchedule(rollNo): ExamEntry[] addExam(courseId, examData): void updateExamRoom(examId, roomNo): void
Related Classes	
ExamSchedule Course Student Admin	

Figure 14.1 - UML Class Diagram

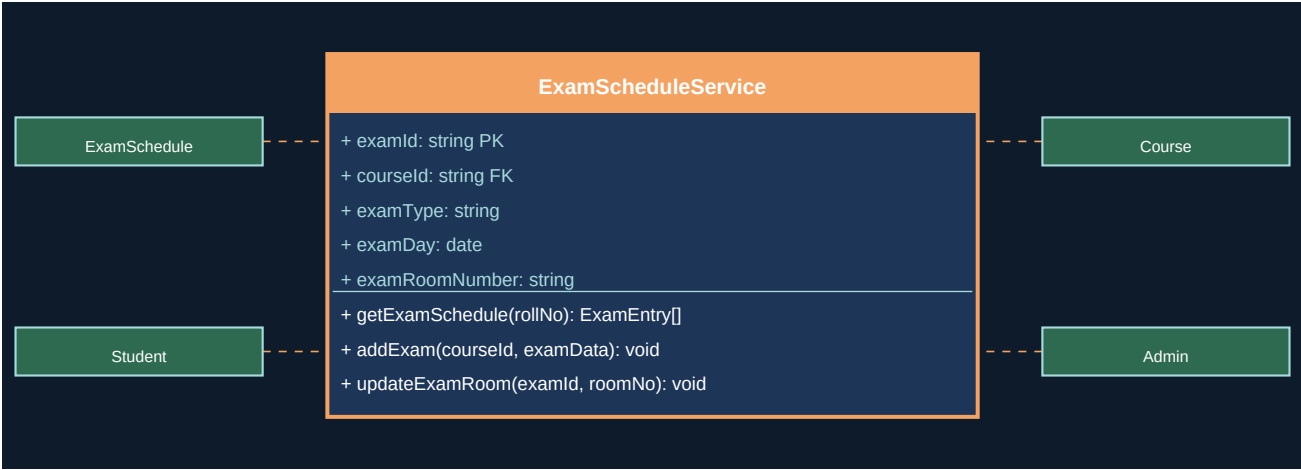
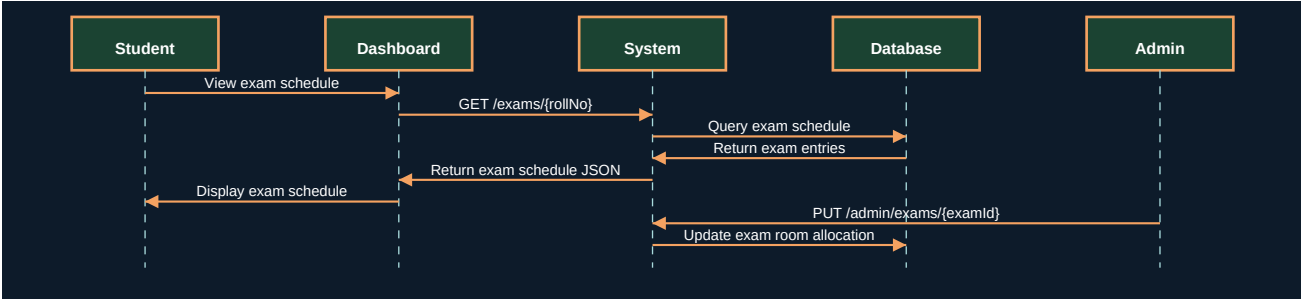


Figure 14.2 - UML Sequence Diagram



Service 15: DocumentVaultService [Nakshatra Kumar]

Description	
Provides secure storage and retrieval for student documents including proofs, certificates, and ID card FIRs. Supports upload, download, and access control per student.	
Attributes	Methods
fileId: string PK fileName: string fileContent: binary requestId: string FK uploadedAt: datetime	uploadDocument(rollNo, file, type): fileId downloadDocument(fileId, rollNo): File listDocuments(rollNo): FileMetadata[] deleteDocument(fileId): void
Related Classes	
Vault CertificateRequest IDCardRequest Student	

Figure 15.1 - UML Class Diagram

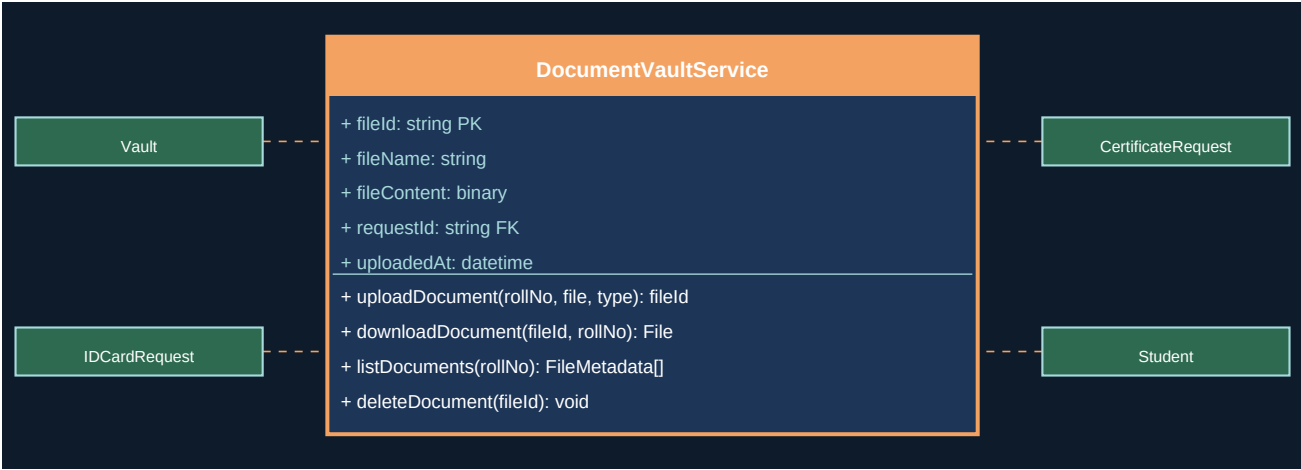
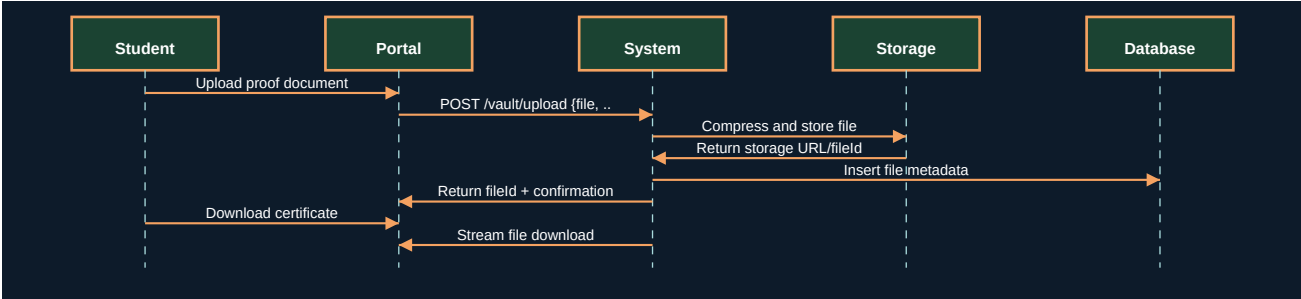


Figure 15.2 - UML Sequence Diagram



Service 16: NotificationService [Tarandeep Kaur]

Description

Sends notifications to students on key academic events: request status changes, timetable updates, exam schedule changes, and certificate readiness via push/email.

Attributes	Methods
notificationId: string PK rollNumber: string FK message: text type: string isRead: boolean sentAt: datetime	sendNotification(rollNo, message, type): void markAsRead(notificationId): void getNotifications(rollNo): Notification[] sendBulk(rollNos, message): void

Related Classes

Notification | Student | Admin

Figure 16.1 - UML Class Diagram

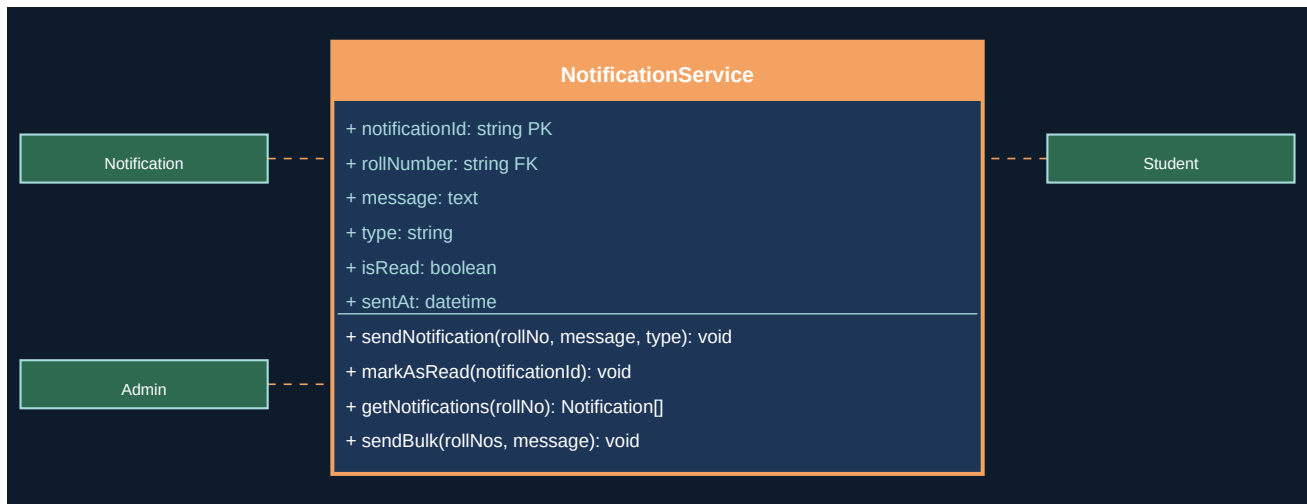
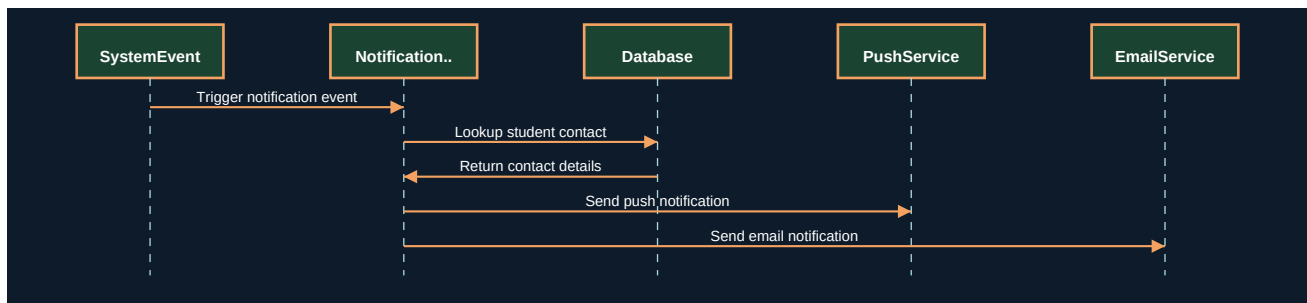


Figure 16.2 - UML Sequence Diagram

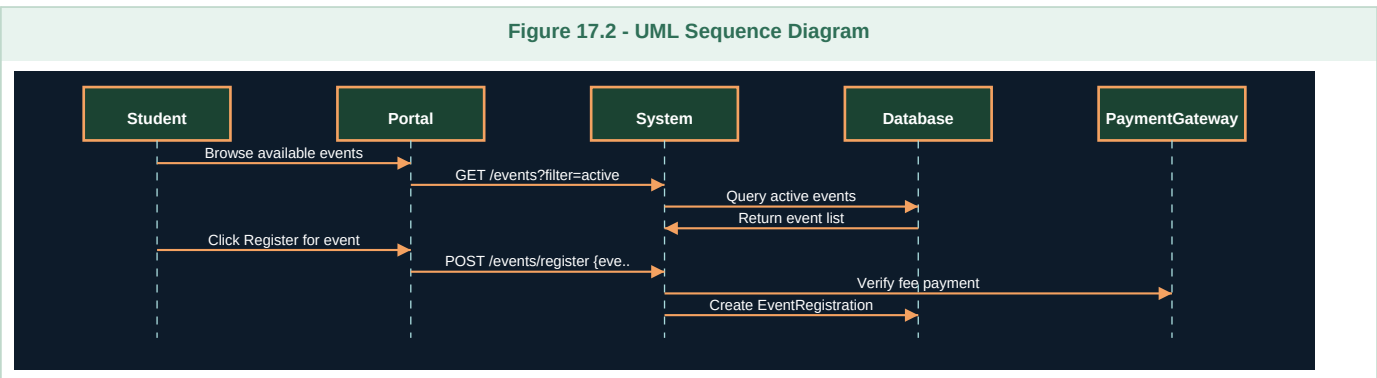
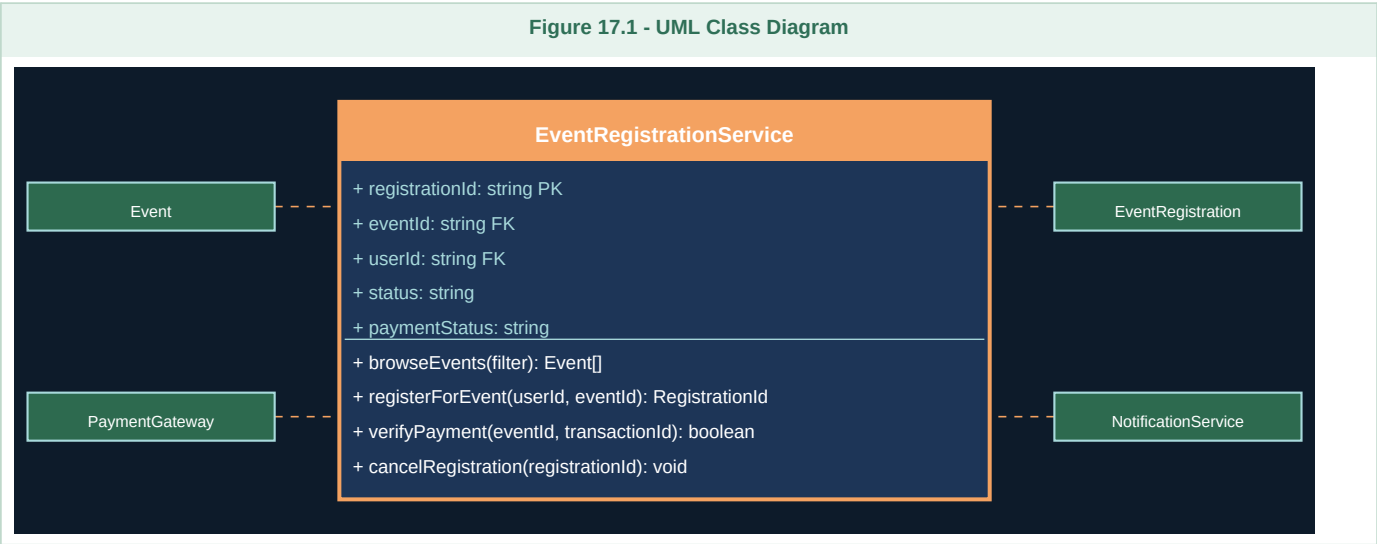


Module 4: Assignment Portal

Members: Botsa Thanmai, Dedeepya Siri Annadevara, Ponga Sai Nikhitha Reddy, Taes Padhiary, Vislavath Bindhu, Pankaj

Service 17: EventRegistrationService [Botsa Thanmai]

Description	
Enables students to browse and register for campus events (cultural, technical, sports, academic). Verifies fee payment via the payment gateway before confirming paid events.	
Attributes	Methods
registrationId: string PK eventId: string FK userId: string FK status: string paymentStatus: string	browseEvents(filter): Event[] registerForEvent(userId, eventId): RegistrationId verifyPayment(eventId, transactionId): boolean cancelRegistration(registrationId): void
Related Classes	
Event EventRegistration PaymentGateway NotificationService	



Service 18: EventPublishingService [Dedeepya Siri Annadevara]

Description
Allows Board Exec to create and publish campus events with title, date, and category. Supports event bookmarking by students and dispatches registration confirmation notifications.

Attributes	Methods
eventId: string PK title: string eventDate: date category: string isFree: boolean status: string	createEvent(title, date, category, isFree): EventId publishEvent(eventId): void bookmarkEvent(userId, eventId): void unpublishEvent(eventId): void
Related Classes	
Event BoardExec NotificationService EventRegistration	

Figure 18.1 - UML Class Diagram

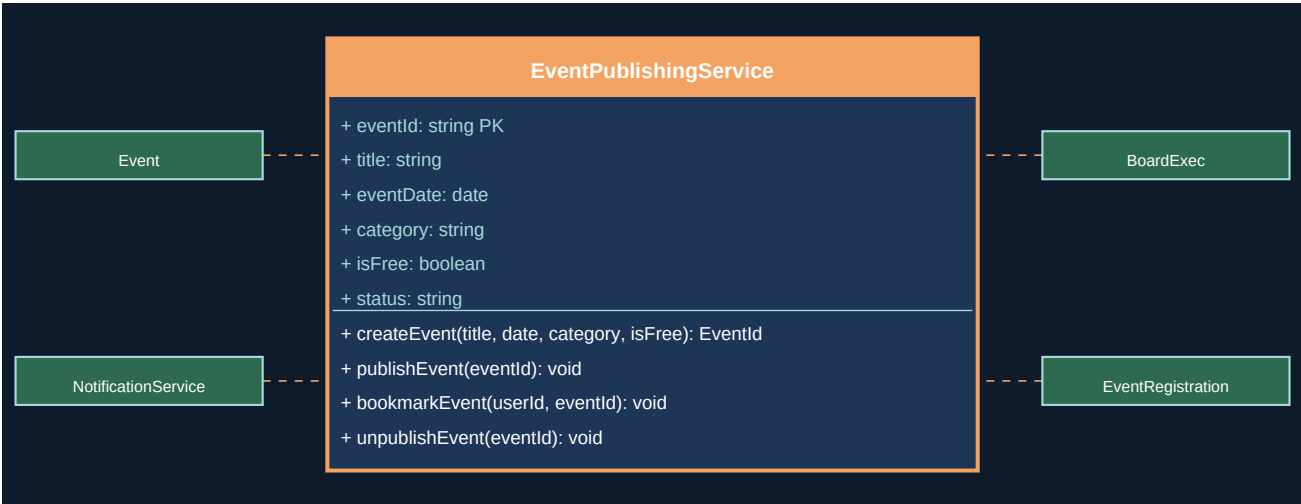
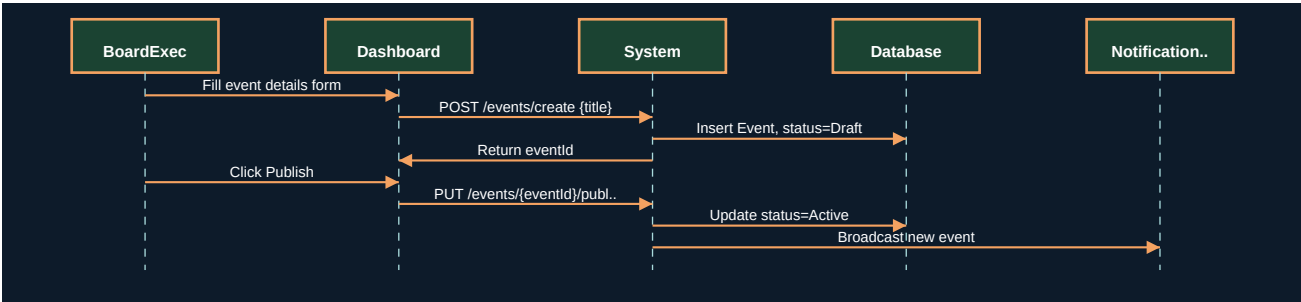


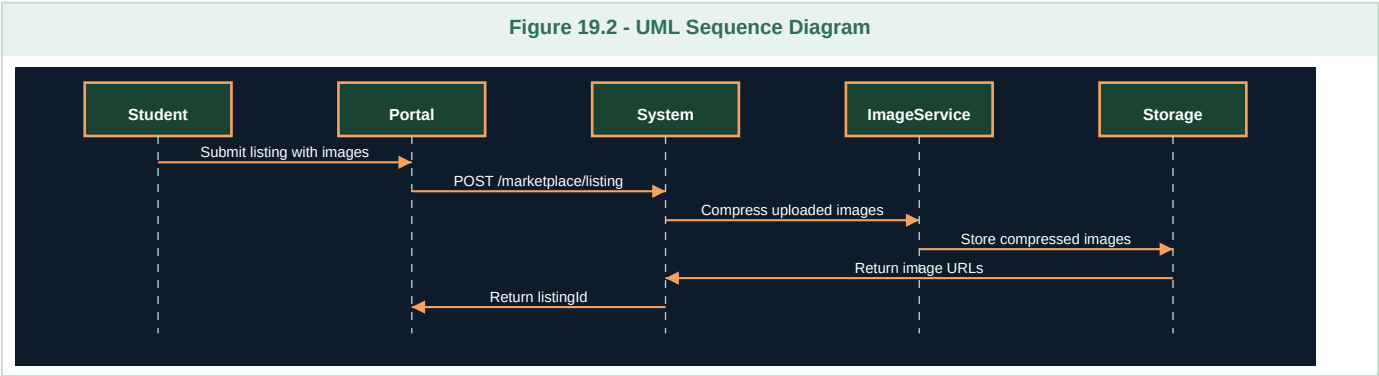
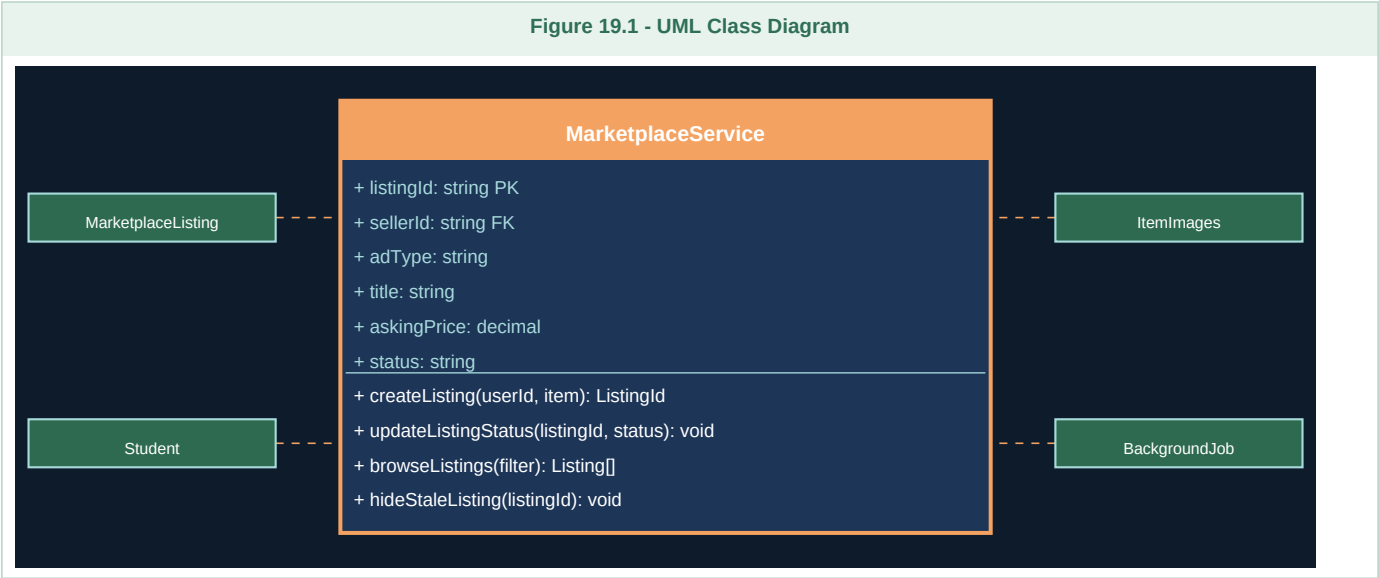
Figure 18.2 - UML Sequence Diagram



Service 19: MarketplaceService [Ponga Sai Nikhitha Reddy]

Description	
Peer-to-peer student marketplace for buying, renting, or giving away items. Supports image upload with compression, status management, and automatic hiding of old listings.	
Attributes	Methods
listingId: string PK sellerId: string FK adType: string title: string askingPrice: decimal status: string	createListing(userId, item): ListingId updateListingStatus(listingId, status): void browseListings(filter): Listing[] hideStaleListing(listingId): void

Related Classes
MarketplaceListing ItemImages Student BackgroundJob



Service 20: ComplaintsPortalService [Taes Padhiary]

Description	
Routes student complaints to the correct channel: IPM (fetches live status from external IPM API) or UPSP/Hostel (submits to internal Maintenance/Admin API with push notifications).	
Attributes	Methods
complaintId: string PK userId: string FK portalType: string category: string status: string	submitIPMComplaint(userId, ipmId): Status submitHostelComplaint(userId, form): ComplaintId getComplaintStatus(complaintId): Status listMyComplaints(userId): Complaint[]
Related Classes	
Complaint IPMPortalAPI MaintenanceAdminAPI NotificationService	

Figure 20.1 - UML Class Diagram

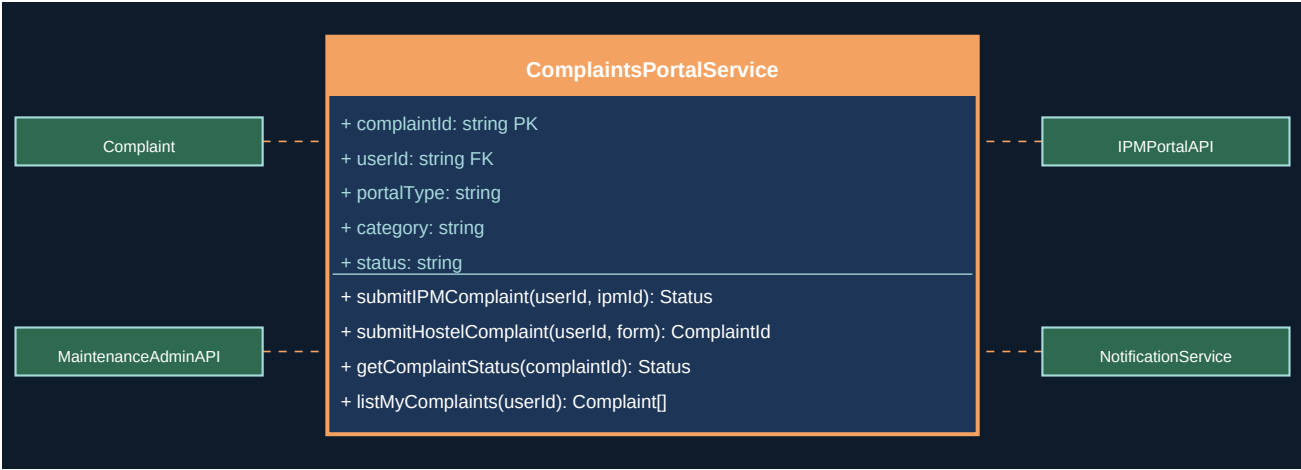
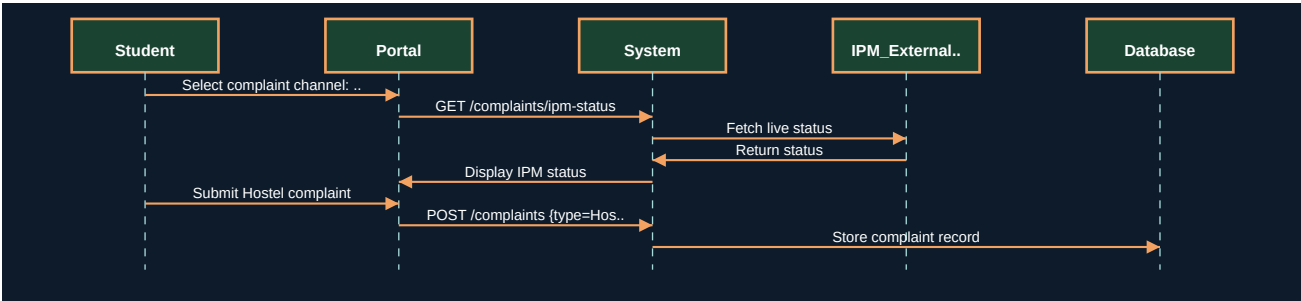


Figure 20.2 - UML Sequence Diagram



Service 21: GateLogQRService [Vislavath Bindhu]

Description

Generates unique encrypted QR codes for student gate entry/exit, encoding student identity, declared destination, and timestamp. Provides a view of historical gate log entries.

Attributes	Methods
logId: int PK userId: int FK gateLocation: string destination: string outTime: timestamp qrToken: string	generateQRCode(userId, destination): QRCode getGateLogs(userId): GateLog[] recordGateEvent(qrData, gateLocation, eventType): void

Related Classes

GateLogs | Users | QREncoder

Figure 21.1 - UML Class Diagram

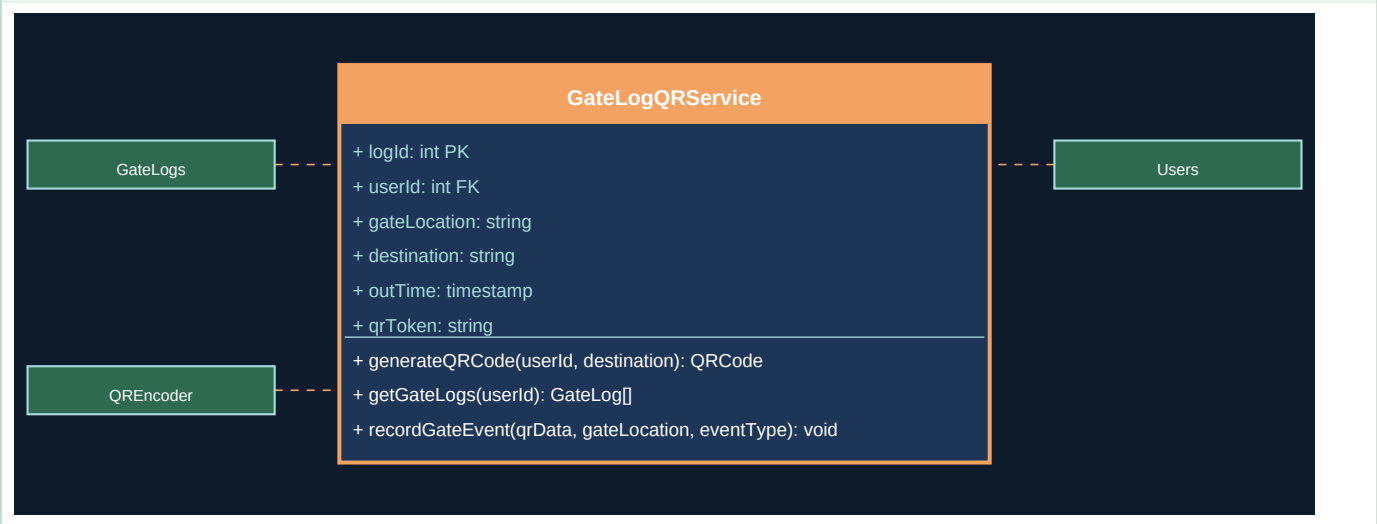
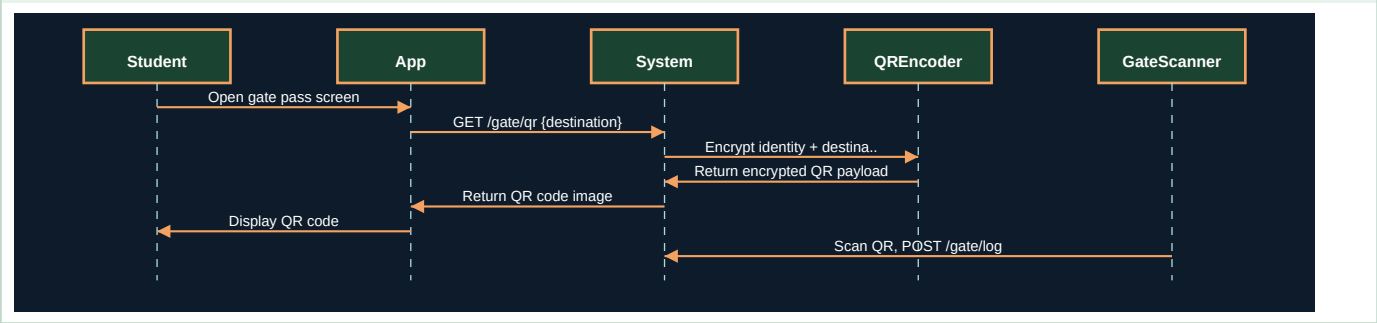


Figure 21.2 - UML Sequence Diagram



Service 22: CabSharingService [Pankaj]

Description

Peer-to-peer cab sharing: a Host posts a ride with source, destination, seats, and time. Other students browse and request to join; the host approves/rejects. Posts auto-archive after pickup time.

Attributes	Methods
shareId: int PK hostId: int FK source: string destination: string availableSeats: int status: string	createCabPost(hostId, source, dest, seats, time): ShareId requestToJoin(shareId, requesterId): RequestId approveRequest(requestId): void archiveExpired(): void

Related Classes

CabShares | CabRequests | Users | NotificationService

Figure 22.1 - UML Class Diagram

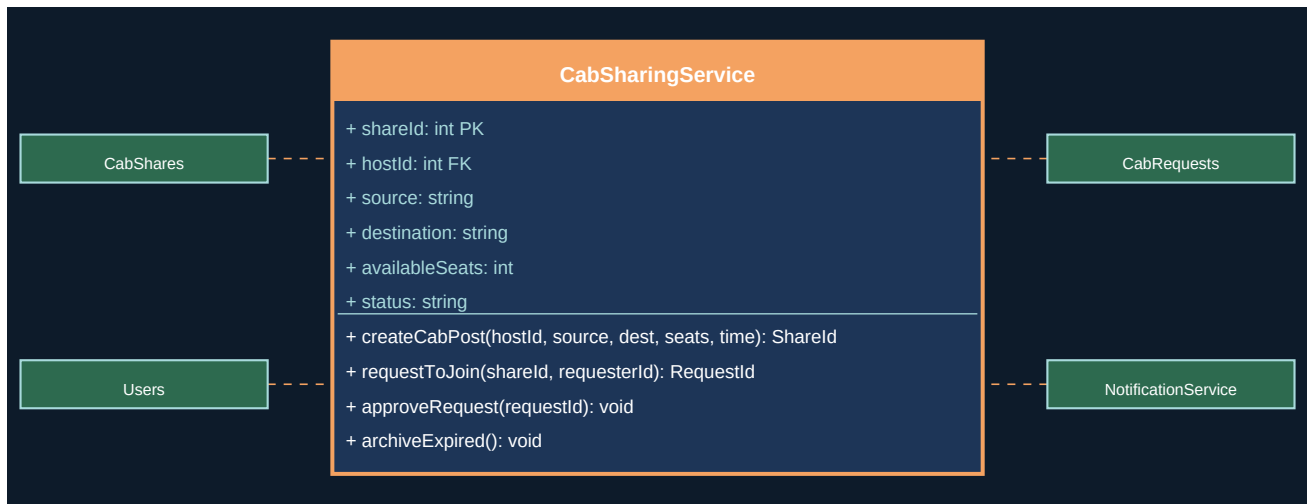
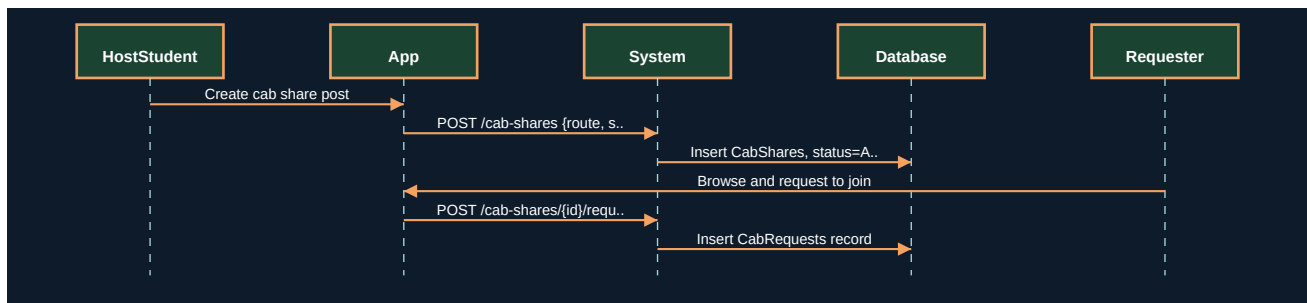


Figure 22.2 - UML Sequence Diagram

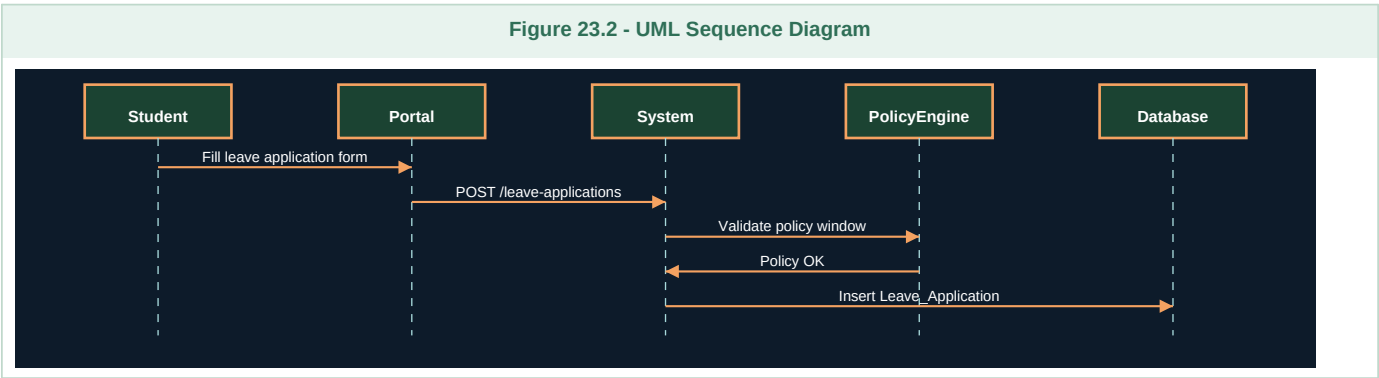
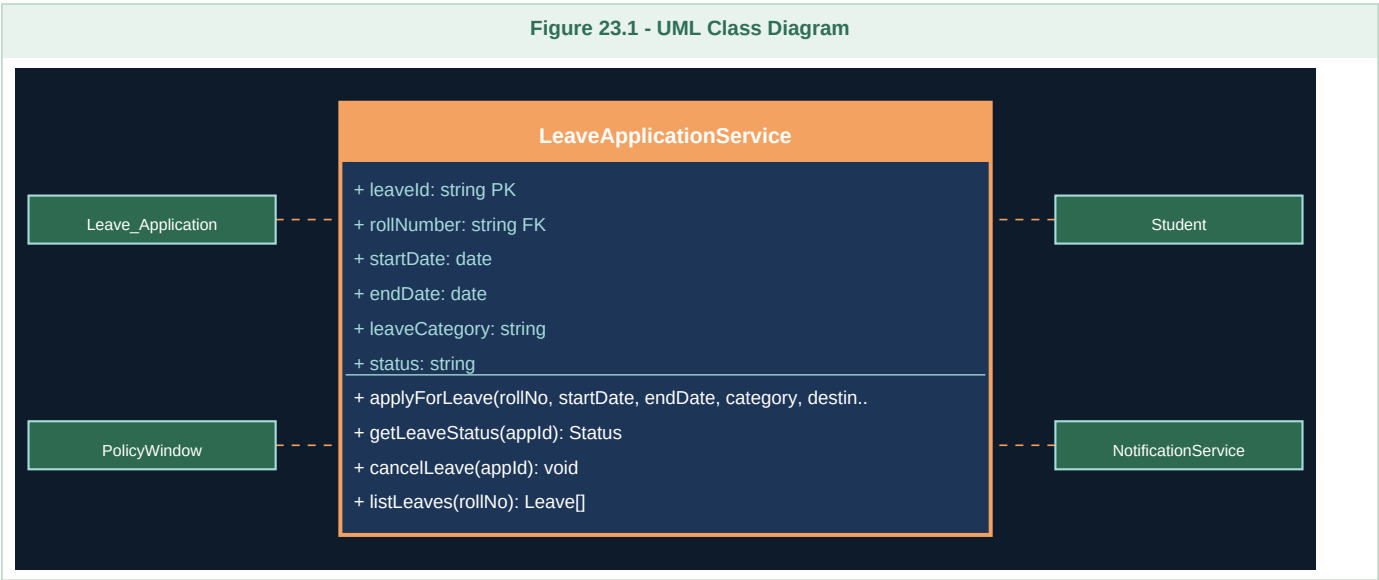


Module 5: Hostel Management

Members: Anubhav Jindal, Biraj Sarkar, Gugulothu Maniprakash, Jash Vaidya, Maneesh S Kolekar, Suram Sai Swapnith Kumar Reddy

Service 23: LeaveApplicationService [Anubhav Jindal]

Description	
Manages student hostel leave applications: validates policy windows, stores the application with dates, destination, and category, notifies Admin/Warden, and tracks approval status.	
Attributes	Methods
leaveld: string PK rollNumber: string FK startDate: date endDate: date leaveCategory: string status: string	applyForLeave(rollNo, startDate, endDate, category, destination): Appld getLeaveStatus(appId): Status cancelLeave(appId): void listLeaves(rollNo): Leave[]
Related Classes	
Leave_Application Student PolicyWindow NotificationService	



Service 24: HostelTransferService [Biraj Sarkar]

Description
Handles hostel room transfer requests: validates no active transfer exists, creates the request with target hostel and reason, and routes it to Admin for approval.

Attributes	Methods
requestId: string PK rollNumber: string FK currentHostelId: string FK targetHostelId: string FK status: string	requestTransfer(rollNo, targetHostelId, reason): RequestId getTransferStatus(requestId): Status cancelTransfer(requestId): void

Related Classes
Transfer_Request Student Hostel Admin

Figure 24.1 - UML Class Diagram

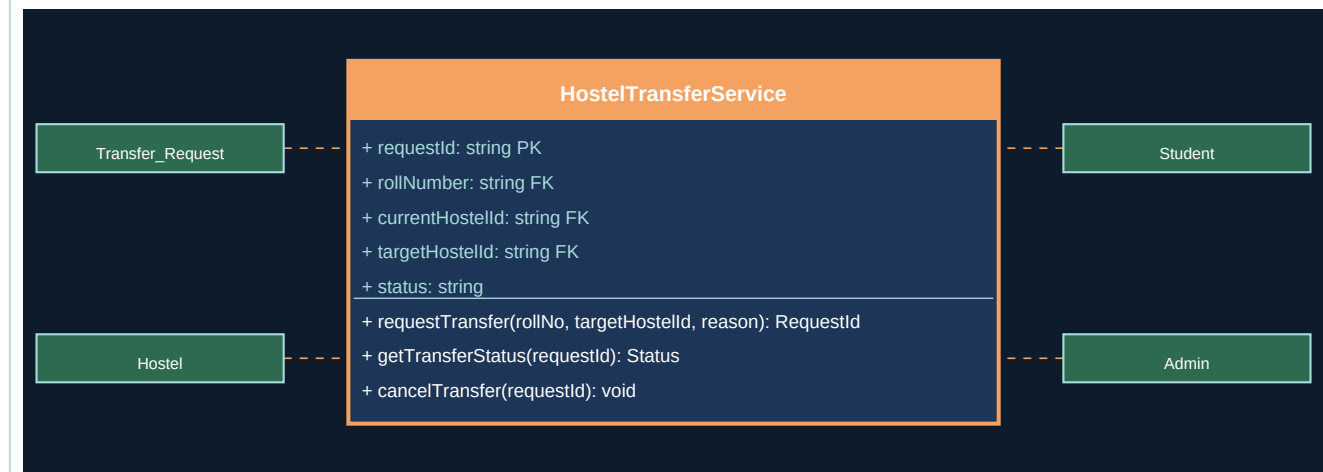
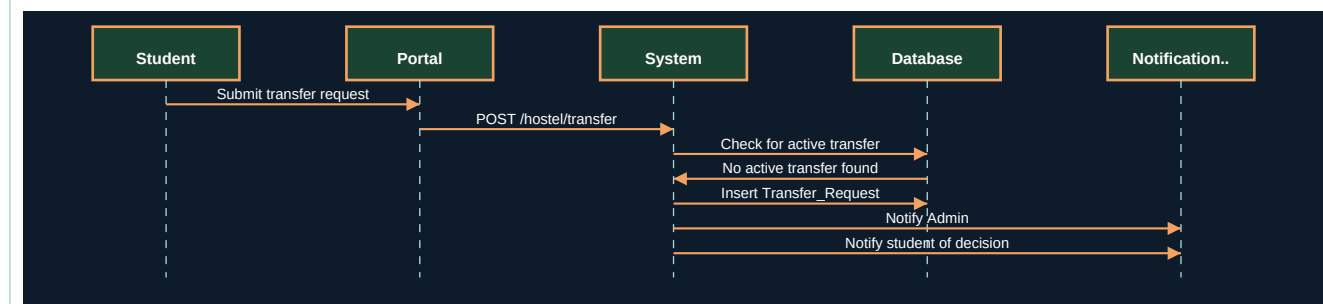


Figure 24.2 - UML Sequence Diagram



Service 25: MessSubscriptionService [Gugulothu Maniprakash]

Description

Allows students to change their mess subscription within policy windows. Deactivates the current subscription and activates a new one; notifies the Mess Secretary.

Attributes	Methods
subscriptionId: string PK rollNumber: string FK messId: string FK startDate: date isActive: boolean	changeMessSubscription(rollNo, newMessId): void getActiveSubscription(rollNo): Subscription getMessList(): Mess[]

Related Classes
Mess_Subscription Mess Student PolicyWindow

Figure 25.1 - UML Class Diagram

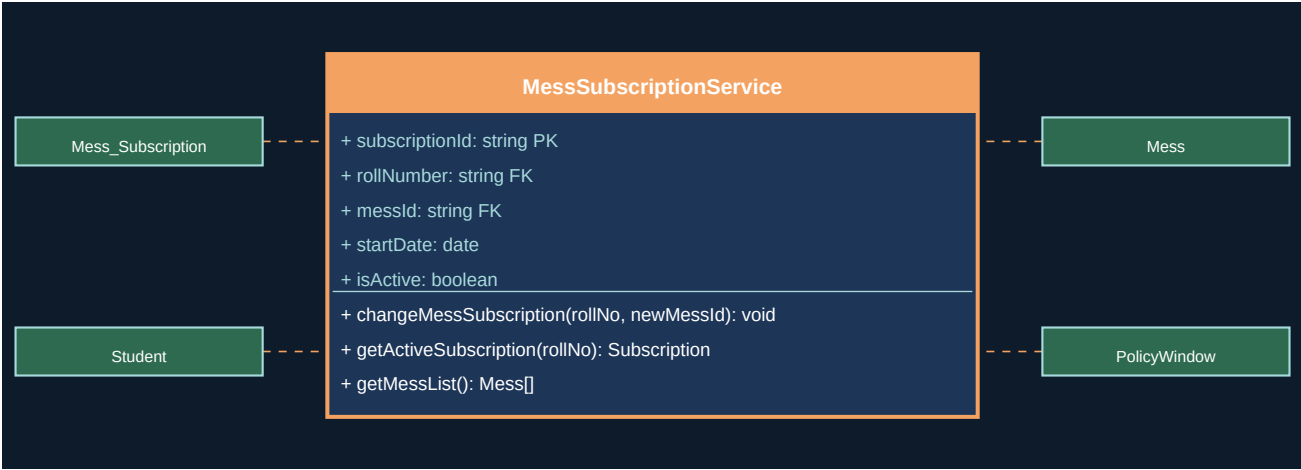
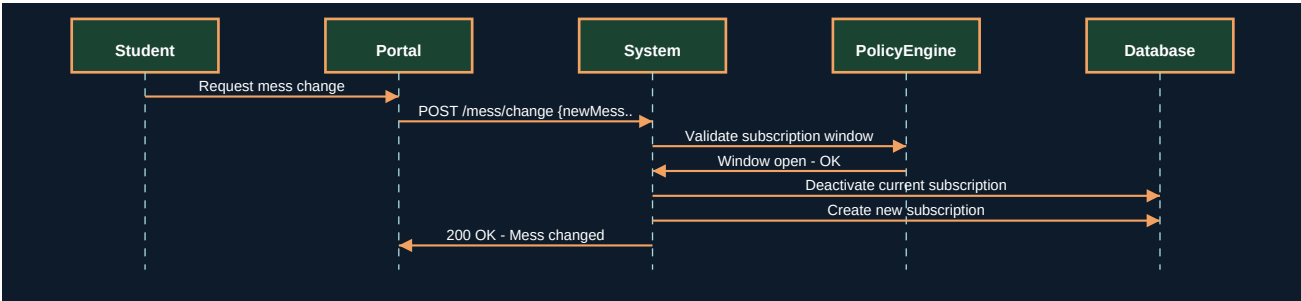


Figure 25.2 - UML Sequence Diagram



Service 26: MessRebateService [Jash Vaidya]

Description	
Processes mess rebate requests linked to approved leave applications. Validates no duplicate rebate claim, creates the request pending Mess Secretary approval.	
Attributes	Methods
rebateId: string PK rollNumber: string FK leaveApplicationId: string FK status: string approvedBy: string	applyForRebate(rollNo, leaveAppId): RebateId getRebateStatus(rebateId): Status approveRebate(rebateId, secId): void
Related Classes	
Mess_Rebate Leave_Application Student MessSecretary	

Figure 26.1 - UML Class Diagram

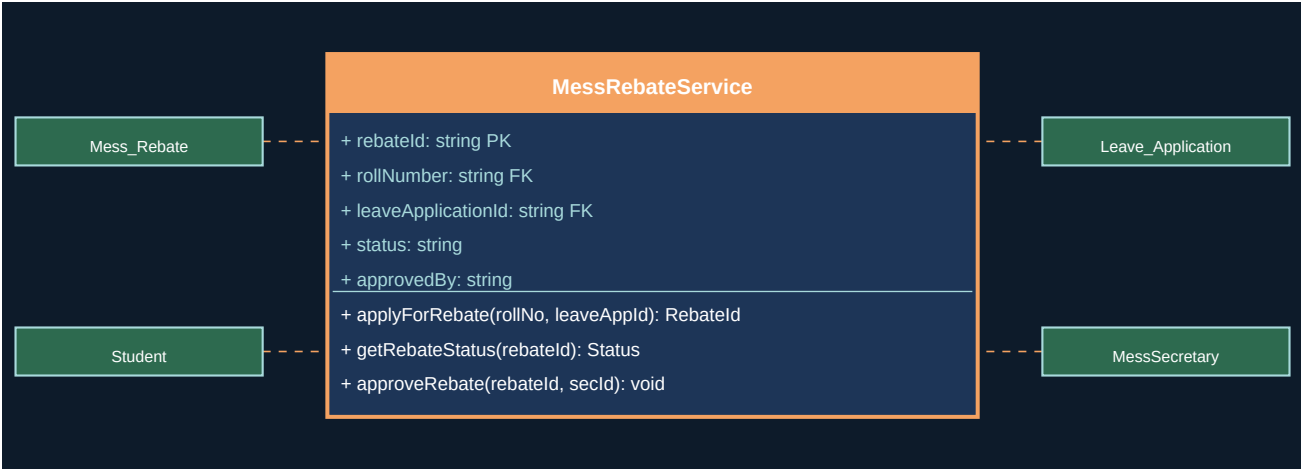
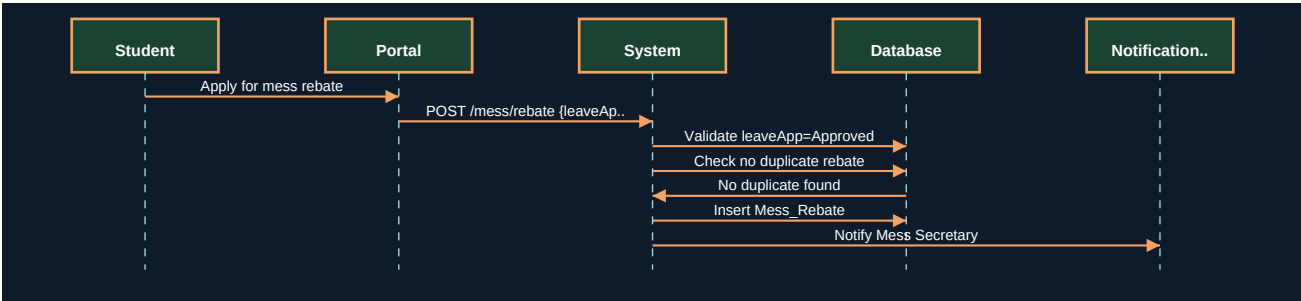


Figure 26.2 - UML Sequence Diagram



Service 27: MessFeedbackService [Maneesh S Kolekar]

Description

Records per-meal ratings and comments for mess food quality. Subscriber feedback carries higher OPI weight. Aggregates ratings to assist OPI computation monthly.

Attributes	Methods
feedbackId: string PK rollNumber: string FK messId: string FK mealRating: int comments: string	submitFeedback(rollNo, messId, meal, rating, comment): void getAggregatedRatings(messId, month): Ratings getOPIScore(messId, month): float

Related Classes

Mess_Feedback | Mess_OPI_Monthly | Mess | MessSecretary

Figure 27.1 - UML Class Diagram

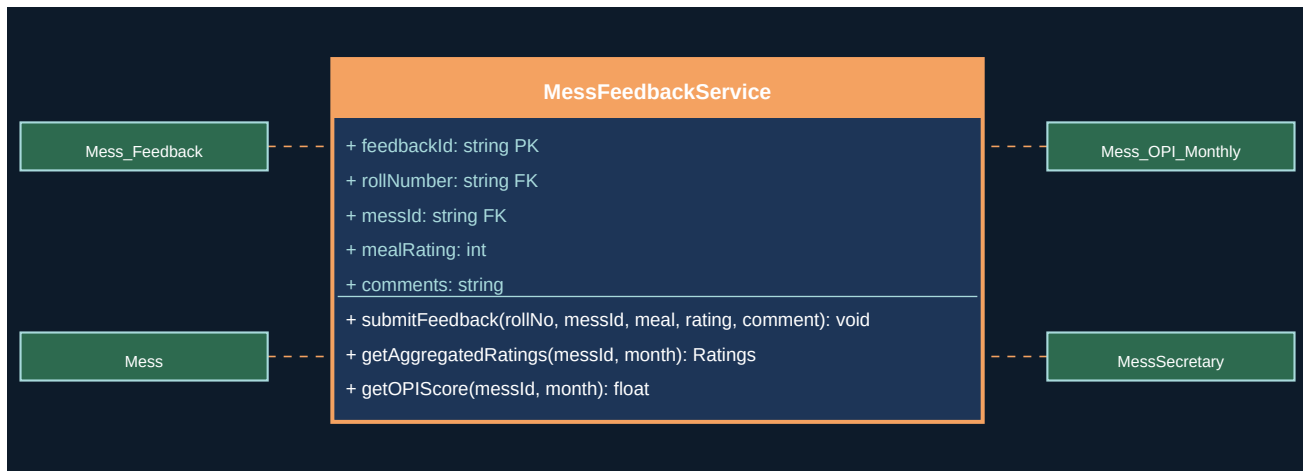
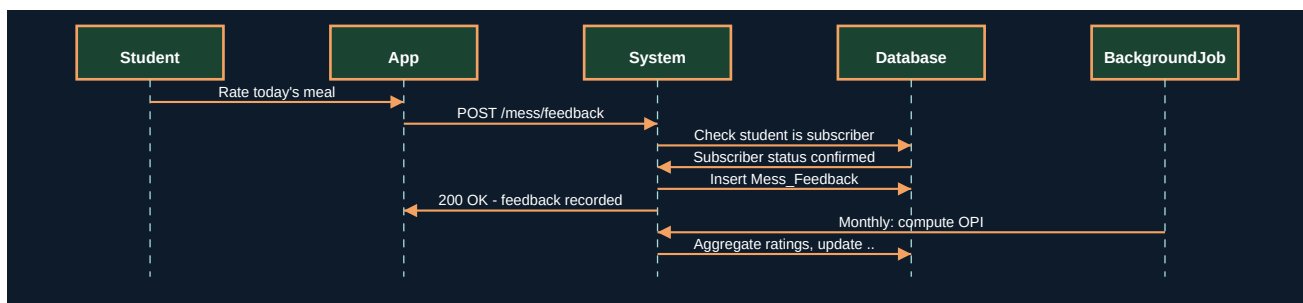


Figure 27.2 - UML Sequence Diagram



Service 28: LaundryQRService [Suram Sai Swapnith Kumar Reddy]

Description

Creates a laundry service request with a unique QR token. Laundry staff scans the QR upon completion, which updates the status to Ready and triggers a push notification to the student.

Attributes

requestId: string PK
 rollNumber: string FK
 qrToken: string
 status: string
 scheduledDate: date

Methods

requestLaundry(rollNo): RequestId
 generateQRToken(requestId): QRCode
 markReady(qrToken): void
 getLaundryStatus(requestId): Status

Related Classes

Service_Request | Student | LaundryStaff | NotificationService

Figure 28.1 - UML Class Diagram

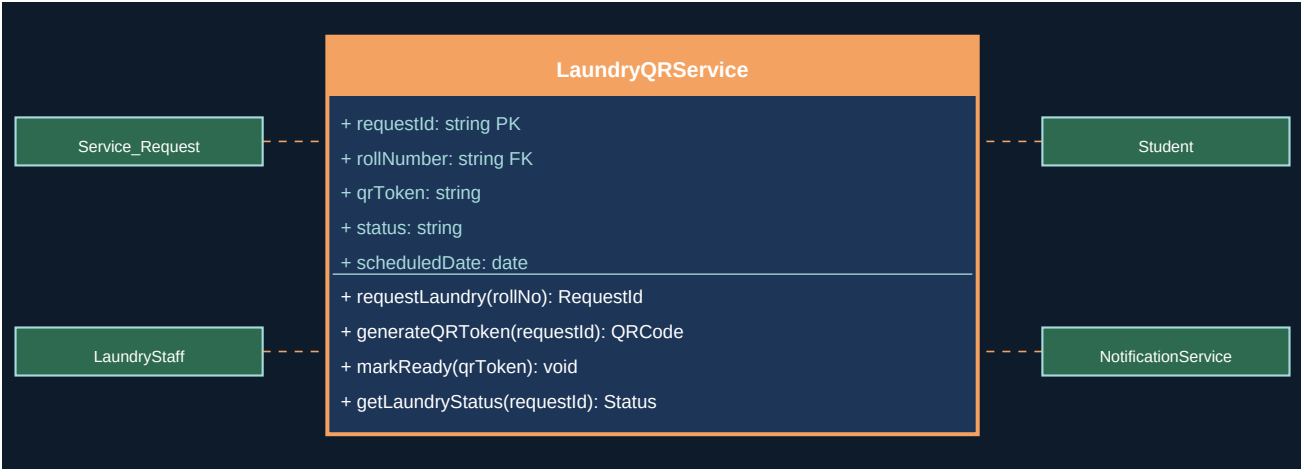
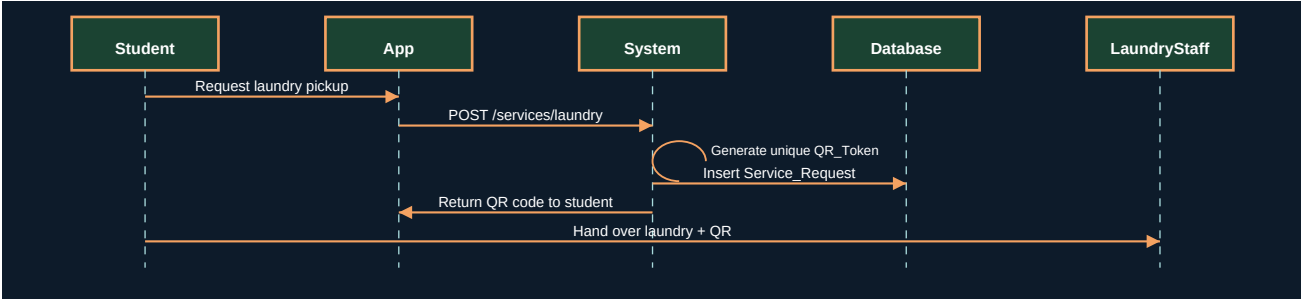


Figure 28.2 - UML Sequence Diagram



Service 29: PolicyWindowService [Module 5 - Policy]

Description

Manages configurable policy windows that control when students can submit leave, subscription changes, or rebate requests. Auto-rejects out-of-window requests with guidance on the next period.

Attributes	Methods
policyId: string PK policyType: string startDate: date endDate: date isActive: boolean	isPolicyWindowOpen(type): boolean getNextWindow(type): PolicyWindow createPolicyWindow(type, start, end): void deactivateWindow(policyId): void

Related Classes

Policy_Window | LeaveApplication | MessSubscription | MessRebate

Figure 29.1 - UML Class Diagram

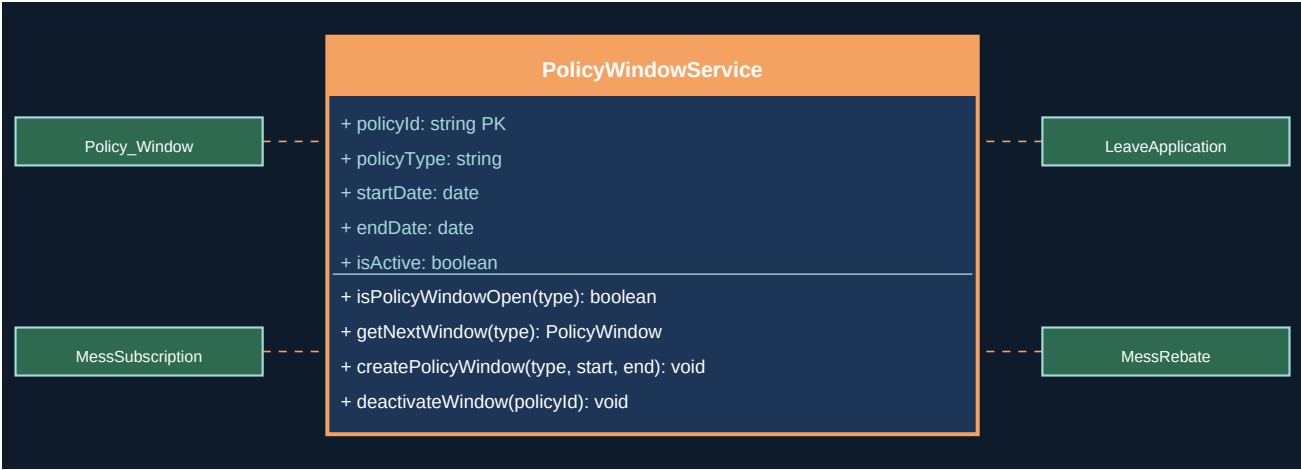
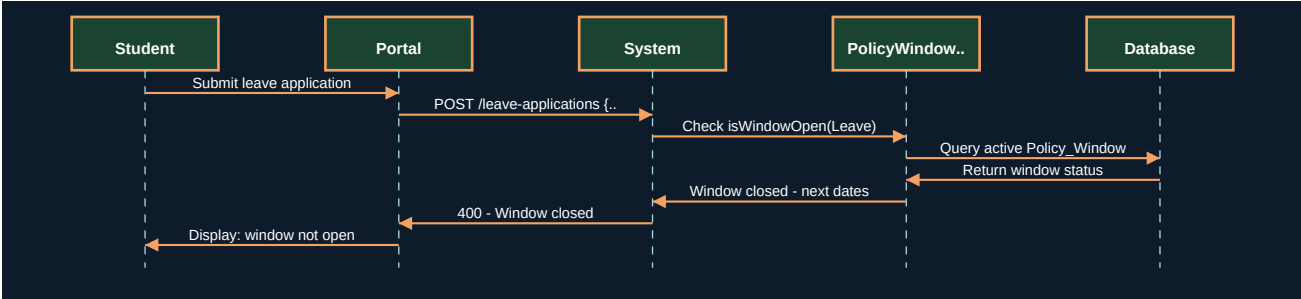


Figure 29.2 - UML Sequence Diagram



Service 30: RoomCleaningService [Module 5 - Cleaning]

Description

Manages student room cleaning service requests, enforcing a free quota per academic period. Rejects requests when quota is exhausted and auto-replenishes at the start of each new period.

Attributes	Methods
requestId: string PK rollNumber: string FK serviceType: string status: string quotaBalance: int	requestCleaning(rollNo): RequestId getQuotaBalance(rollNo): int deductQuota(rollNo): void replenishQuota(rollNo, period): void

Related Classes

Service_Request | Student | PolicyWindow | QuotaTracker

Figure 30.1 - UML Class Diagram

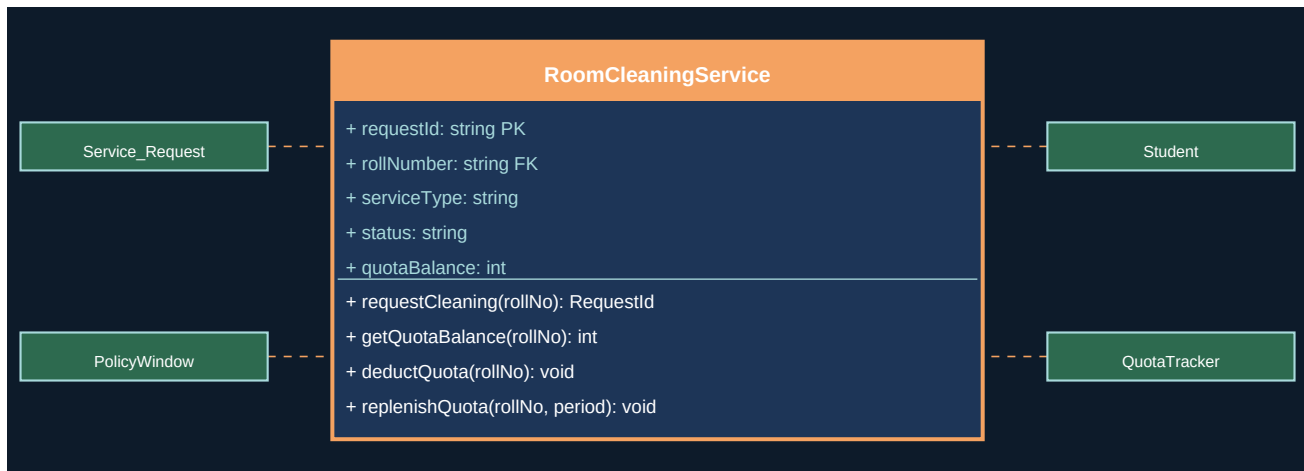
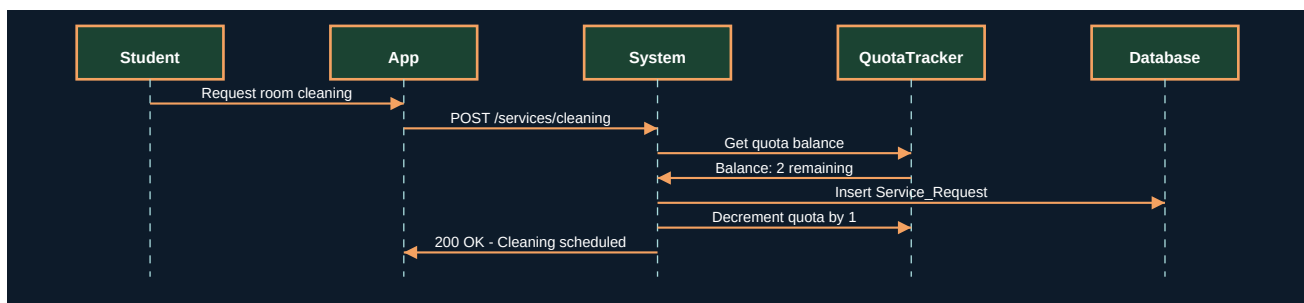


Figure 30.2 - UML Sequence Diagram

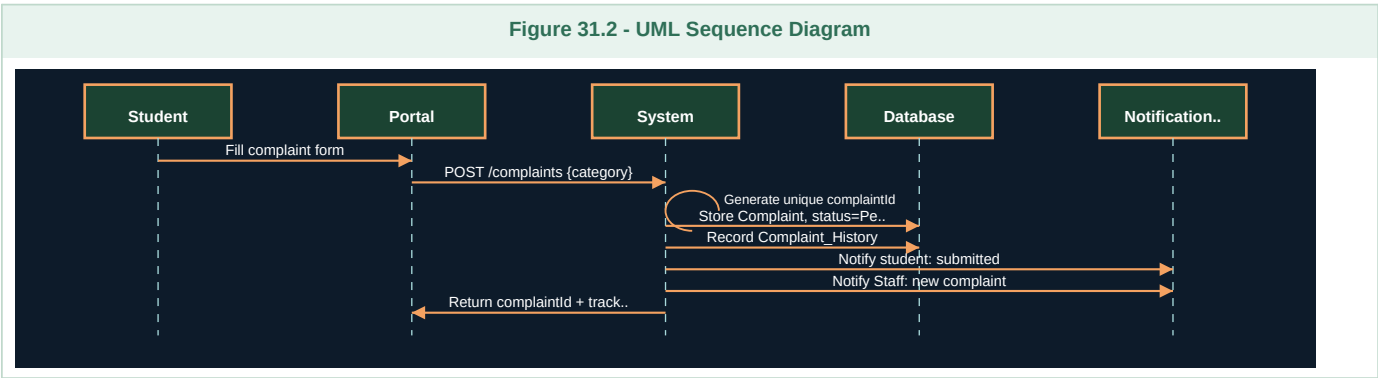
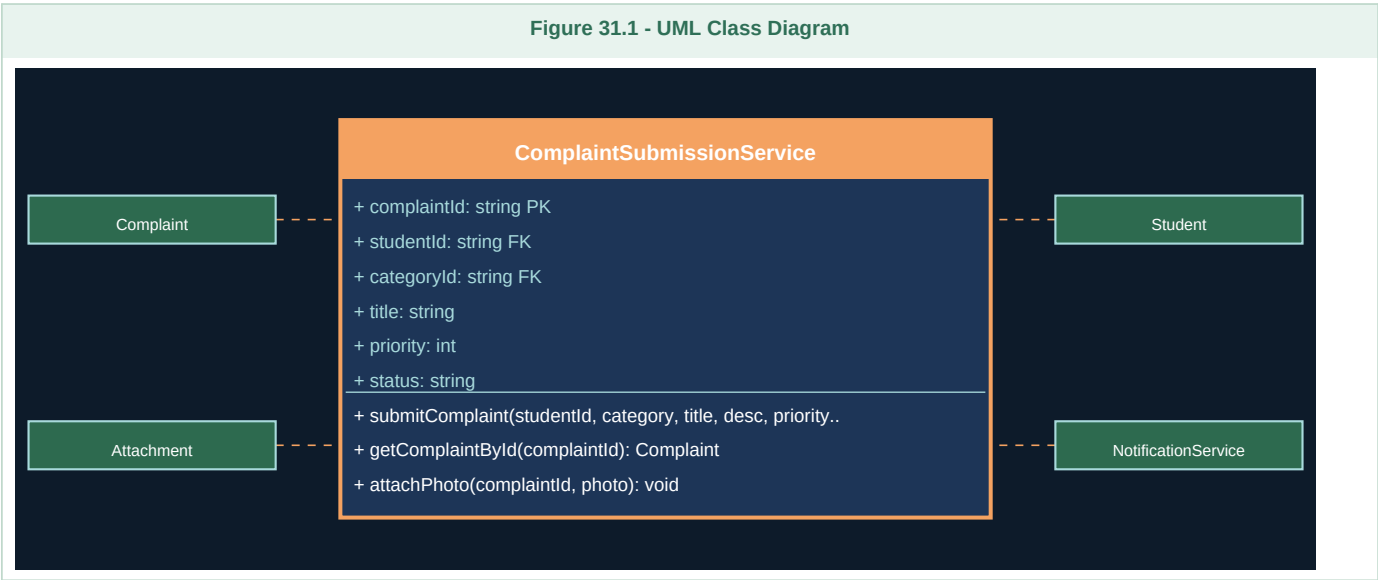


Module 6: Complaint System

Members: Ashish Kumar, Kusireddy Sri Durga Siva Ganesh, Prateek, Aryan

Service 31: ComplaintSubmissionService [Ashish Kumar]

Description	
Allows students to submit complaints with category, title, description, priority, and optional photo attachment. Generates a unique complaint ID and records creation timestamp.	
Attributes	Methods
complaintId: string PK studentId: string FK categoryId: string FK title: string priority: int status: string	submitComplaint(studentId, category, title, desc, priority): ComplaintId getComplaintById(complaintId): Complaint attachPhoto(complaintId, photo): void
Related Classes	
Complaint Student Attachment NotificationService	



Service 32: ComplaintAssignmentService [Kusireddy Sri Durga Siva Ganesh]

Description
Assigns complaints to staff members either manually by admin or automatically based on category and workload. Notifies the assigned staff and records the assignment.

Attributes	Methods
complaintId: string FK assignedStaffId: string FK assignedBy: string assignedAt: datetime	assignToStaff(complaintId, staffId, adminId): void autoAssign(complaintId): staffId getWorkload(staffId): int reassign(complaintId, newStaffId): void
Related Classes	
Complaint Staff Admin NotificationService	

Figure 32.1 - UML Class Diagram

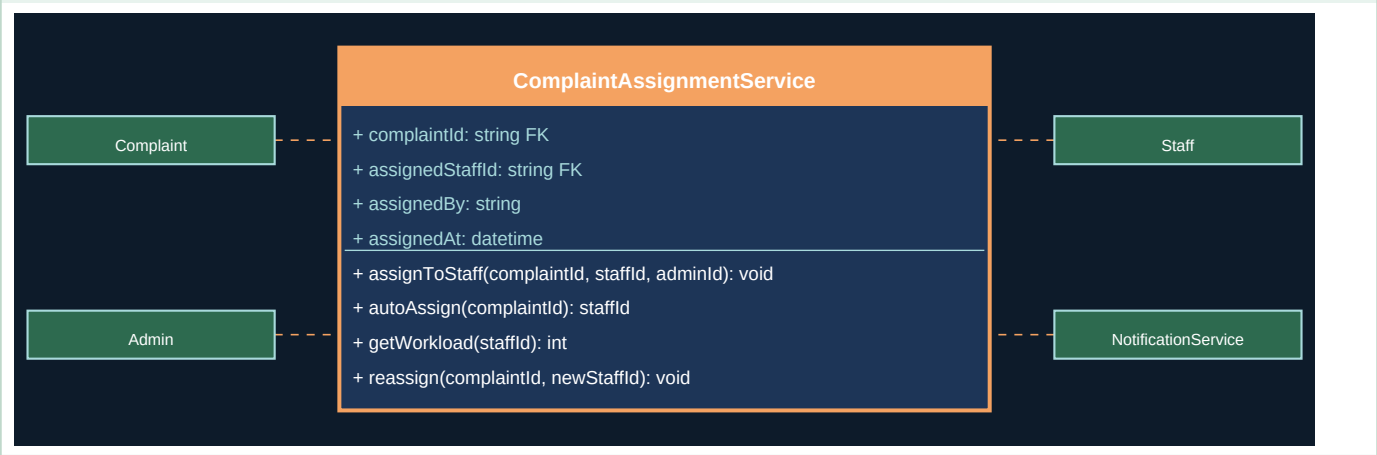
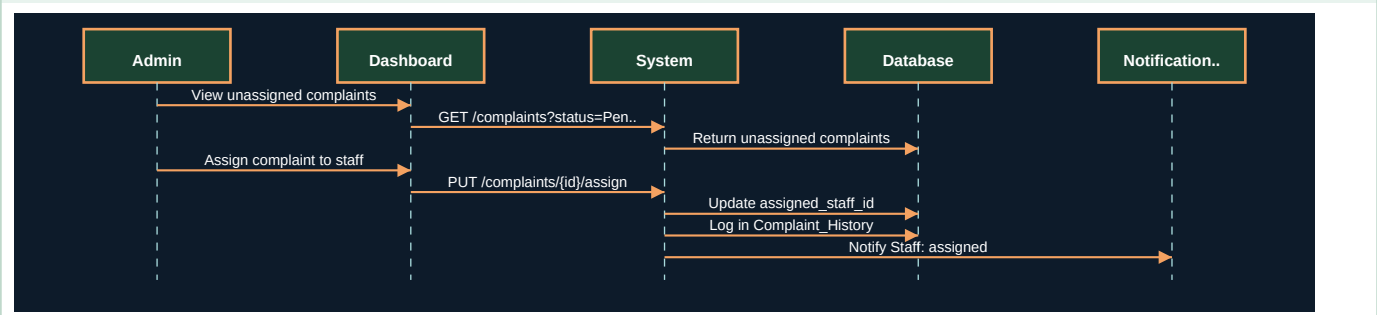


Figure 32.2 - UML Sequence Diagram



Service 33: ComplaintTrackingService [Prateek]

Description	
Enables students to view full complaint status, history of status changes, staff comments, and timeline. Ensures complete traceability through Complaint_History logging.	
Attributes	Methods
historyId: string PK complaintId: string FK oldStatusId: int newStatusId: int changedBy: string changedAt: datetime	getComplaintStatus(complaintId): Status getComplaintHistory(complaintId): History[] getComments(complaintId): Comment[] viewTimeline(complaintId): Timeline
Related Classes	
Complaint Complaint_History Comment Student	

Figure 33.1 - UML Class Diagram

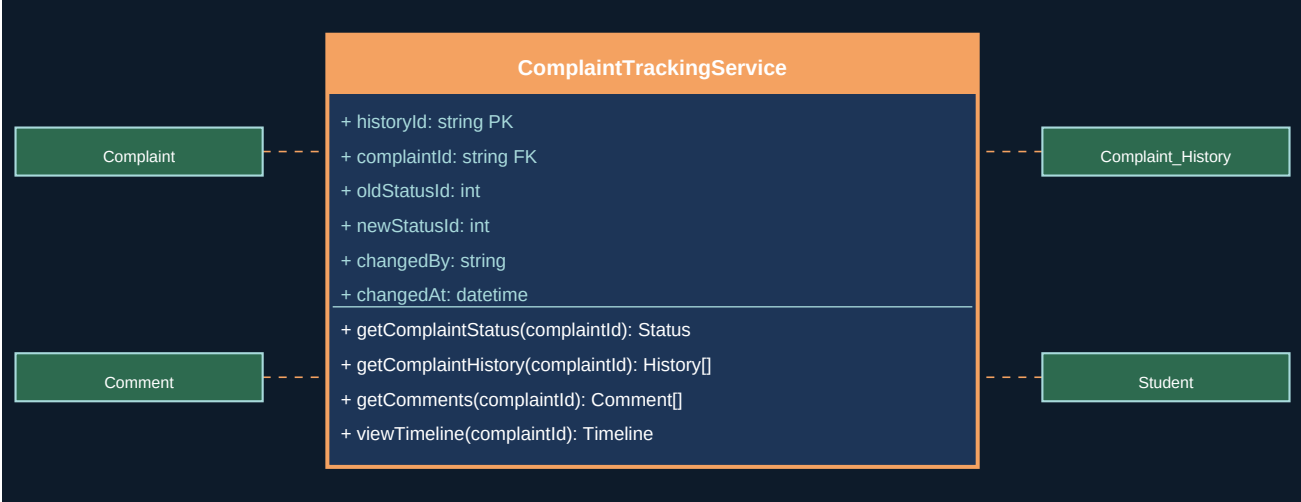
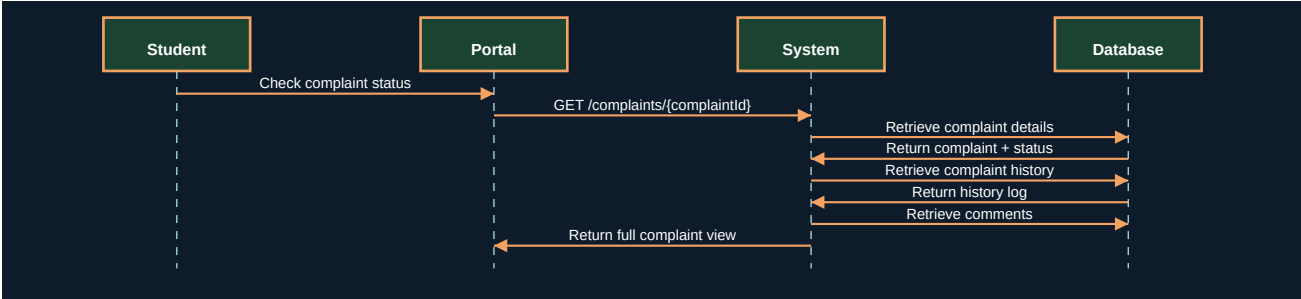


Figure 33.2 - UML Sequence Diagram



Service 34: ComplaintStatusUpdateService [Aryan]

Description

Allows staff to update complaint status through workflow stages (Pending to In Progress to Resolved). Records every status change in Complaint_History for full auditability.

Attributes

complaintId: string FK
statusId: int
statusName: string
changedBy: string
changedAt: datetime

Methods

updateStatus(complaintId, newStatus, staffId): void
resolveComplaint(complaintId, staffId, note): void
requestMoreInfo(complaintId, message): void

Related Classes

Complaint | Complaint_Status | Complaint_History | Staff

Figure 34.1 - UML Class Diagram

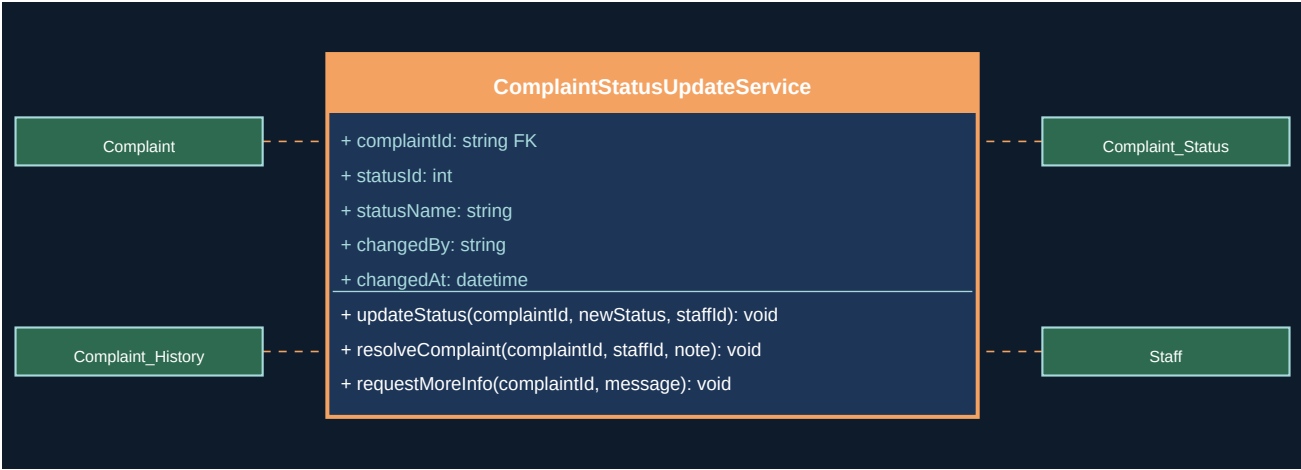
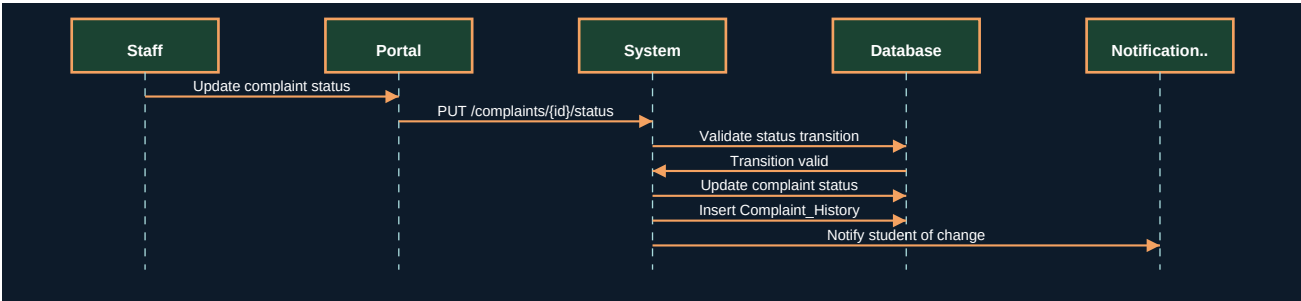


Figure 34.2 - UML Sequence Diagram



Service 35: FeedbackService [Ashish Kumar (M6)]

Description

Enables students to provide post-resolution feedback with numerical ratings and comments. Feedback data feeds into service quality dashboards for admin review.

Attributes	Methods
feedbackId: string PK complaintId: string FK rating: int comments: text createdAt: datetime	submitFeedback(complaintId, rating, comments): FeedbackId getFeedbackForComplaint(complaintId): Feedback getAverageRating(staffId): float

Related Classes

Feedback | Complaint | Student | Staff

Figure 35.1 - UML Class Diagram

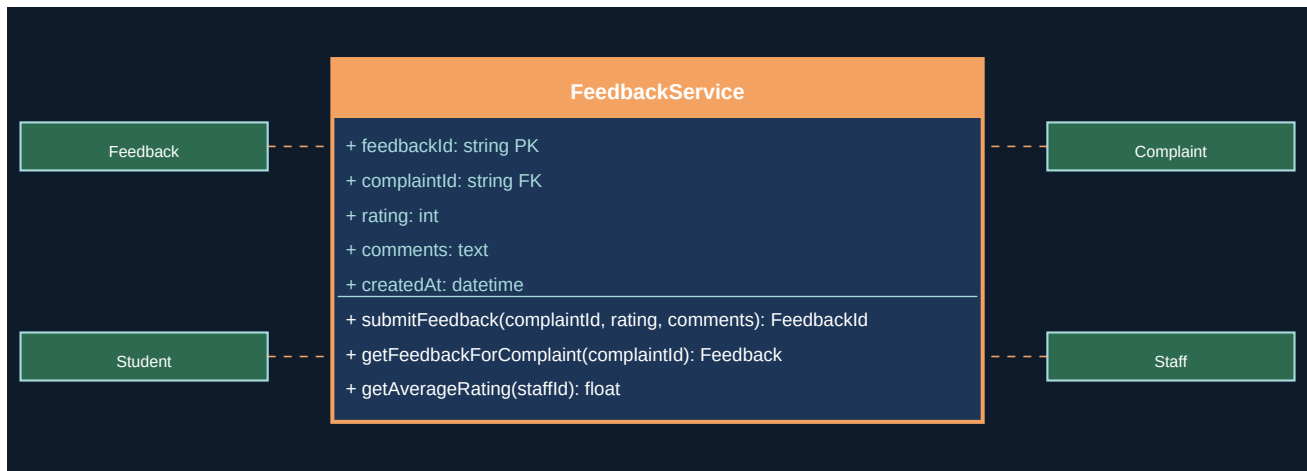
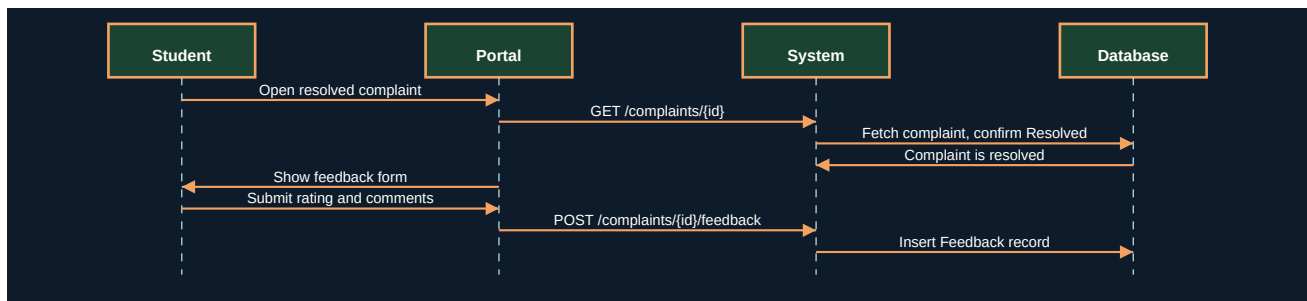


Figure 35.2 - UML Sequence Diagram



Testing Team

Members: chokkaku venkata yaswanth, praveen kumar kanchari, Sachin Kumar Tanti, Shravya Lanka, Shrey Chandra

Service 36: IntegrationTestService [chokkaku venkata yaswanth]

Description

Provides automated integration testing across all portal modules, validating API endpoints, database operations, and cross-module interactions. Reports test results with pass/fail metrics.

Attributes	Methods
testId: string PK moduleName: string endpoint: string expectedStatus: int passed: boolean	runIntegrationTests(module): TestReport validateAPIContracts(endpointList): ValidationResult generateTestReport(suite): Report

Related Classes

TestSuite | APIContract | Database | AllModules

Figure 36.1 - UML Class Diagram

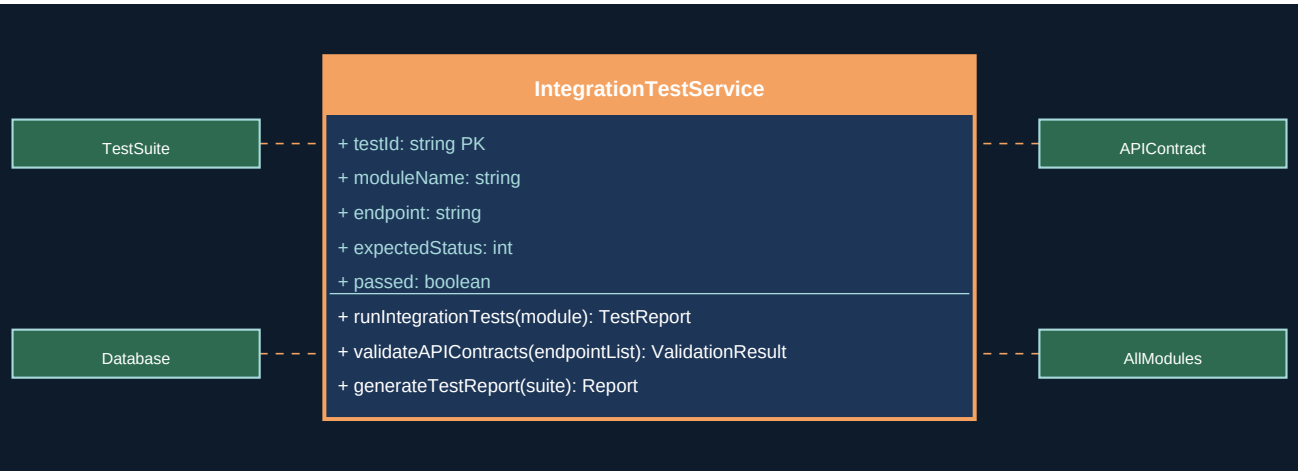
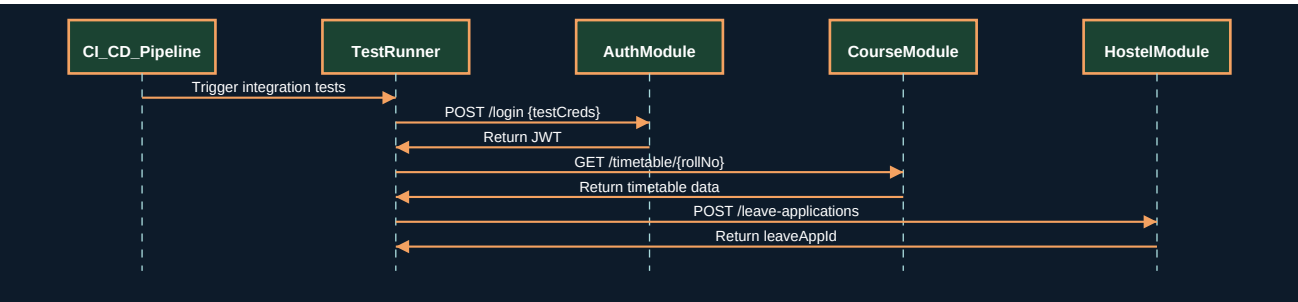


Figure 36.2 - UML Sequence Diagram



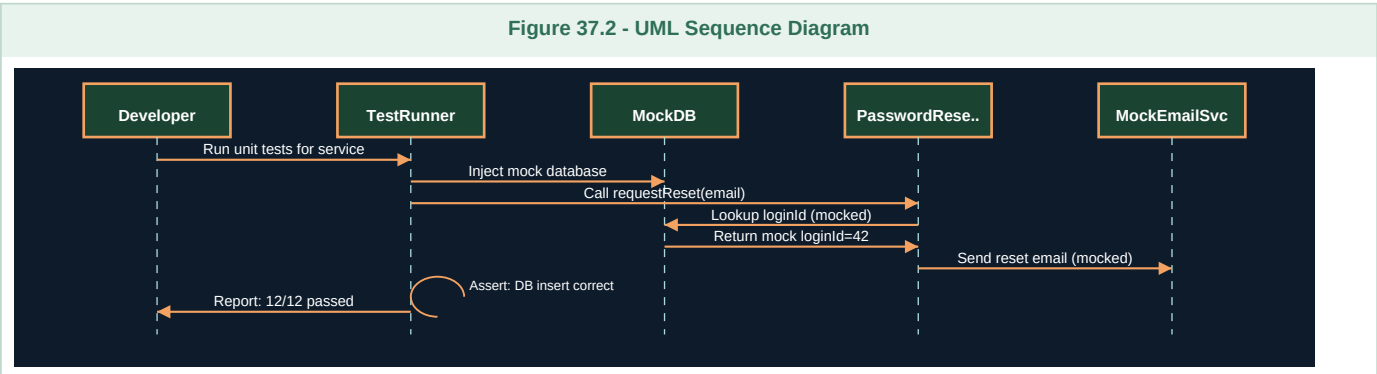
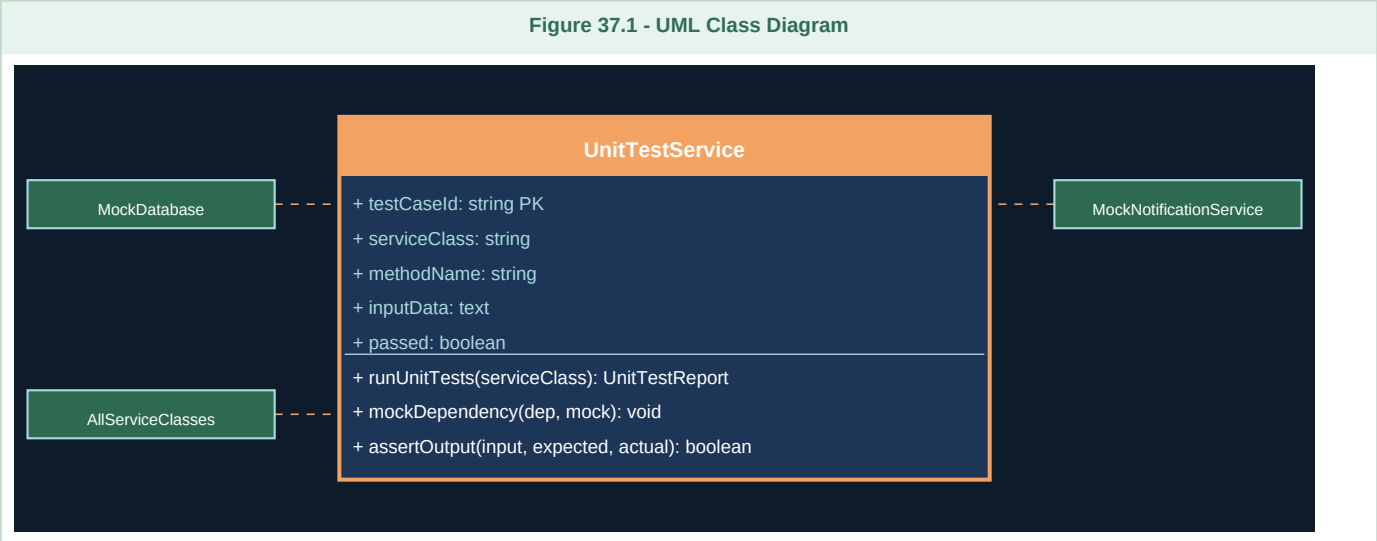
Service 37: UnitTestService [praveen kumar kanchari]

Description

Executes unit tests for individual service functions, validating business logic, input validation, edge cases, and error handling in isolation from external dependencies.

Attributes	Methods
testCaselId: string PK serviceClass: string methodName: string inputData: text passed: boolean	runUnitTests(serviceClass): UnitTestReport mockDependency(dep, mock): void assertOutput(input, expected, actual): boolean

Related Classes
MockDatabase MockNotificationService AllServiceClasses



Service 38: PerformanceTestService [Sachin Kumar Tanti]

Description
Conducts load and stress testing on critical portal endpoints to verify system performance under concurrent user loads, measuring response times and identifying bottlenecks.

Attributes	Methods
testRunId: string PK endpoint: string concurrentUsers: int avgResponseMs: float errorRate: float	runLoadTest(endpoint, concurrentUsers, duration): PerformanceReport measureResponseTime(endpoint): Metrics identifyBottlenecks(report): Bottleneck[]

Related Classes
LoadTestSuite MetricsCollector AllAPIs

Figure 38.1 - UML Class Diagram

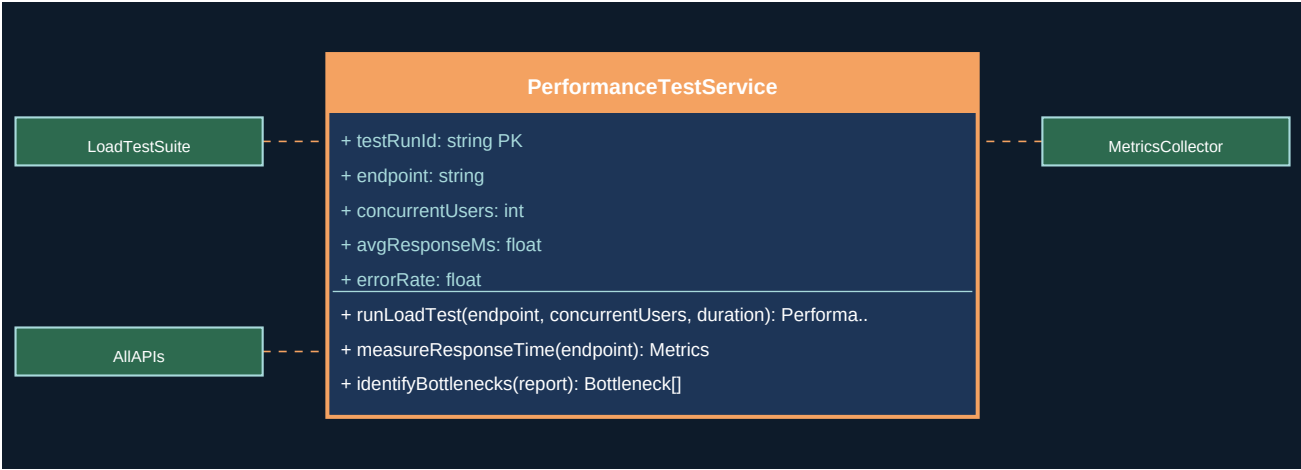
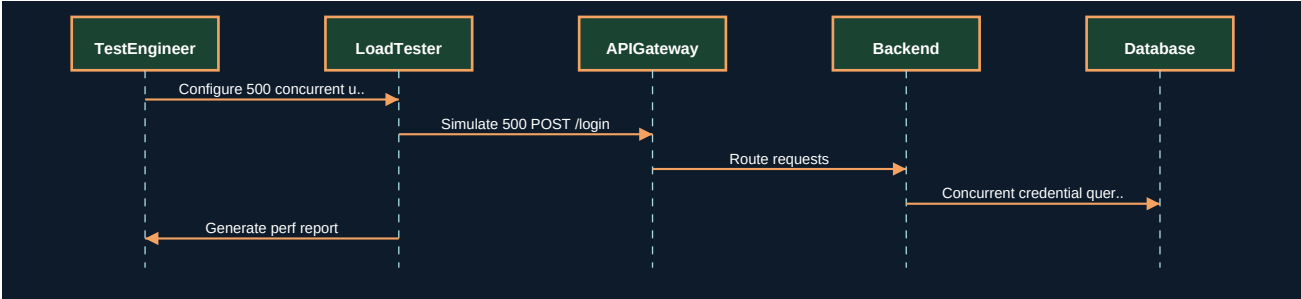


Figure 38.2 - UML Sequence Diagram



Service 39: SecurityTestService [Shravya Lanka]

Description

Validates security controls across the portal including authentication bypasses, SQL injection, RBAC violations, JWT tampering, and session fixation vulnerabilities.

Attributes	Methods
testId: string PK vulnerability: string endpoint: string severity: string passed: boolean	testAuthBypass(endpoint): SecurityResult testSQLInjection(endpoint, payload): SecurityResult testRBACViolation(token, resource): SecurityResult testJWTTampering(token): SecurityResult

Related Classes

AuthModule | RBACService | SessionManagement | AllAPIs

Figure 39.1 - UML Class Diagram

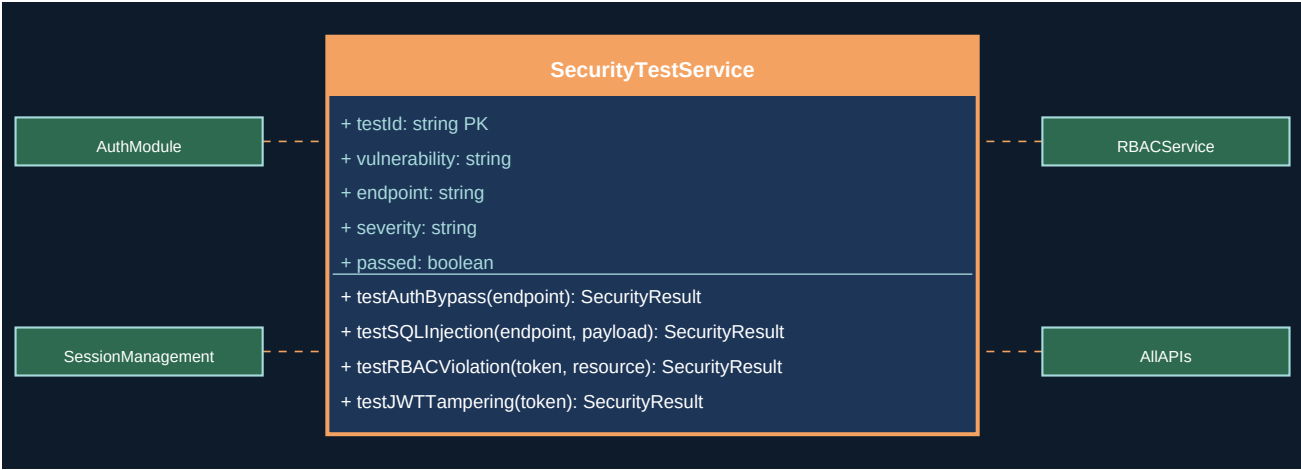
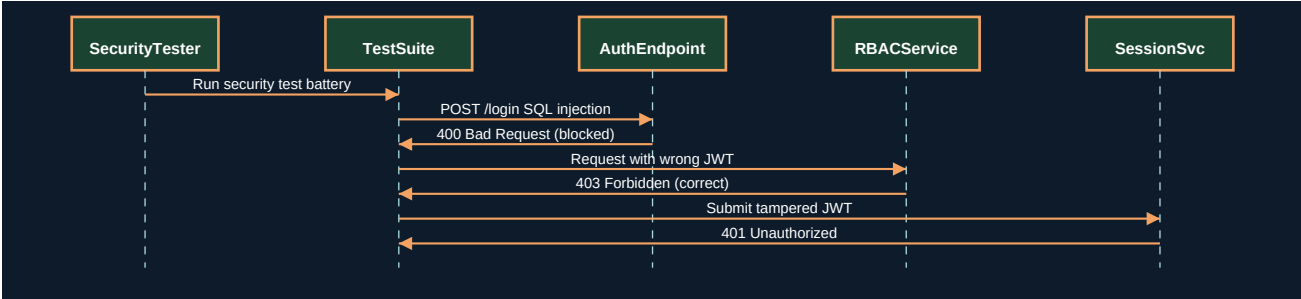


Figure 39.2 - UML Sequence Diagram



Service 40: RegressionTestService [Shrey Chandra]

Description

Runs full regression test suites after every code change to ensure existing functionality remains unbroken. Tracks test pass rates over time and alerts on failures.

Attributes	Methods
runId: string PK commitId: string totalTests: int passed: int passRate: float timestamp: datetime	runRegression(commitId): RegressionReport compareWithBaseline(report, baseline): Delta alertOnFailure(failures): void

Related Classes

TestSuite | CI_CDPipeline | AllModules | AlertService

Figure 40.1 - UML Class Diagram

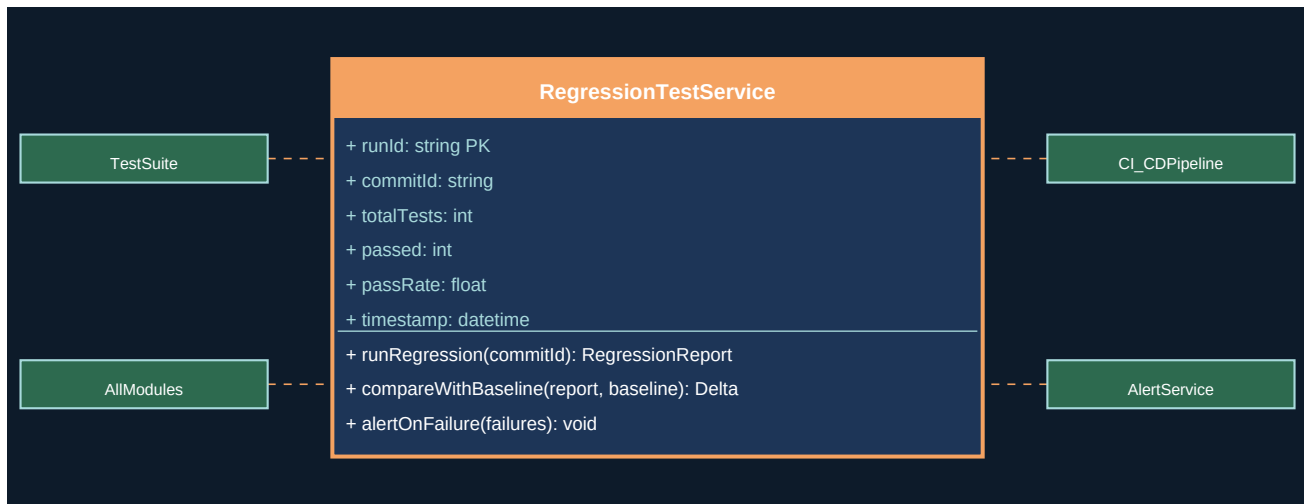


Figure 40.2 - UML Sequence Diagram

